



Event-aware analysis of cross-city visitor flows using large language models and social media data

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ABSTRACT

Understanding the fluctuations of cross-city travel demand is crucial for the efficient operation of intercity transport systems. Public events, such as concerts and fireworks displays, can cause irregular surges in cross-city travel demand, leading to potential overcrowding, travel delays, and public safety concerns. To better anticipate and accommodate such demand surges, it is essential to estimate cross-city visitor flows with awareness of public events. While prior studies primarily focused on the impacts of a single mega event or disruptions around an individual venue, this study extends the scope by proposing a generalizable framework to analyze visitor flows under multiple concurrent events. We propose to leverage large language models (LLMs) to extract event features from multi-source online information and massive user-generated content on social media platforms. Specifically, social media popularity metrics are designed to capture the effects of online promotion and word-of-mouth in attracting visitors, with their effectiveness verified through comparative analysis. An event-aware machine learning model is then adopted to uncover the specific impacts of different event features and ultimately predict visitor flows for upcoming events. Using real-world events, social media, and visitor arrival data from Hong Kong, the framework is applied to predict daily flows of Chinese Mainland visitors, achieving a testing R-squared of over 85%. We further investigate the heterogeneous event impacts on visitor numbers across different event types and major travel modes. Both promotional popularity and word-of-mouth popularity are found to be associated with increased visitor flows, but the specific effects vary by the event type. This association is more pronounced among visitors arriving by metro and high-speed rail, while it has less effect on air travelers. Through a case study of a specific venue, we demonstrate how event-aware visitor flow prediction facilitates coordinated inter-agency measures and guides the development of specialized transport policies. Such analysis enables the implementation of targeted policies, including responsive operations of border control, dedicated shuttle services to event venues, and comprehensive on-site traffic management strategies.

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1. Introduction

Understanding cross-city travel demand is crucial for the efficient operation of intercity transport infrastructure, including airports, railway stations, highways, and shuttle transit systems. However, such travel demand often fluctuates irregularly due to various external factors, such as extreme weather conditions (Lu et al., 2014; Yang et al., 2021) and holidays (Li et al., 2020). In recent years, public events, such as concerts and fireworks displays, have become an increasingly significant driver of cross-city travel demand surges. Policymakers actively leverage frequent and diverse events to attract visitors and boost local economy (Pike and Page, 2014). The combination of widespread event promotion on social media and improved intercity transport connectivity further encourages people to travel across cities to attend a wide range of events (Zhao et al., 2019; Wang et al., 2018; Zeng and Gerritsen, 2014; Xiang and Gretzel, 2010; Li et al., 2020). Existing studies examining the impact of public events on travel demand are relatively limited and often focus on a specific event venue using a single data source (Chen et al., 2020; Liang et al., 2024; Pereira et al., 2015; Rodrigues et al., 2018). Some researchers have considered public events as disruptive factors affecting visitor arrivals (Heller and Stephenson, 2021; Zhang et al., 2024), but usually with a sole focus on occasional international mega events or specific sports games. The joint effect of diverse and concurrent events on cross-city visitor flows remains a largely unexplored research area.

It is well recognized that popular events can draw large crowds, leading to unexpected congestion and overcrowding at cross-city transport facilities, local public transit systems, and specific event sites. Timely insights about cross-city visitor flows are critical to facilitate coordinated measures across government agencies and guide specialized transport policies. Such information enables operators of cross-city transport facilities to identify key entry points susceptible to these disruptions and adjust the allocation of personnel and resources accordingly to minimize delays. Local public transit operators should offer shuttle services to event venues, while also scaling up operations by increasing service frequency and extending operating hours. At specific event sites, transport authorities should collaborate closely with traffic police, emergency services, parking facility managers, and event organizers to implement comprehensive on-site traffic and crowd management strategies. However, local authorities often lack a systematic approach to estimate potential visitor flows attracted by these diverse events in advance, due to several factors: (1) event information is fragmented across multiple online sources, as events are organized by different entities; (2) while many events are ticketed, visitors do not necessarily book them prior to the event day, and there are certain events, such as fireworks displays, that do not involve ticket sales; and (3) multiple events can occur on the same day, which may lead to synergetic effects on cross-city visitor flows. Therefore, to effectively manage such irregular demand surges, it is necessary to develop a systematic event-aware analytical framework for dynamic prediction of cross-city visitor flows under diverse and concurrent events.

In today's digital age, social media data serves as a valuable resource for estimating the potential popularity of events. On the one hand, these platforms play a pivotal role in event marketing (Hu et al., 2022) before events take place. On the other hand, participants often share their experiences on social media after attending events, creating word-of-mouth endorsements (Hu et al., 2022; Cui et al., 2018; Duan et al., 2008; Dellarocas et al., 2007). Both pre-event promotion and post-event experience sharing can significantly affect individuals' intentions to attend events (Li et al., 2023; Hudson et al., 2015), and user engagements with these social media posts can serve as an indicator for the events' potential to attract visitors in the future (Boivin and Tanguay, 2019; Hudson et al., 2015; Zeng and Gerritsen, 2014; Xiang and Gretzel, 2010; Dedeoğlu et al., 2020). Therefore, social media popularity of events may be incorporated as a predictive factor for visitor flows (Zhang et al., 2024). However, quantifying such popularity requires extracting meaningful and relevant information from noisy data. Previous studies on short-term demand predictions often focused on the number of social media check-ins as an indicator of user attendance within subsequent hours (Ni et al., 2016; Xue et al., 2022). In these cases, relevant posts can be filtered directly based on their timestamps and geo-tags. However, effective transport service adjustments (e.g., special shuttle services) often require visitor flow prediction at least a few days before the event. It is thus essential to capture early social media posts ranging from event promotions to post-event experiences; identifying the relevance of these posts requires a semantic evaluation of their textual content. Existing studies often manually screen promotional posts from official accounts (Pereira et al., 2015), overlooking user-generated content that shares personal event experiences. To systematically extract social media popularity from both promotional and experience-sharing posts, more efficient and reliable natural language processing techniques are needed.

To address these challenges, this study presents an event-aware analytical framework to extract event features from online and social media data and then utilize these features for cross-city visitor flow prediction. First, we compile comprehensive event data across various categories and scales in a unified data structure. To incorporate social media popularity as an indicator of potential event appeal, we propose to leverage large language models (LLMs) to process multi-source event information online and analyze the semantics of massive user-generated content on social media platforms. A Gradient Boosted Decision Trees (GBDT) rolling prediction model is then adapted to incorporate these event features and predict visitor flows for upcoming events. The proposed framework is demonstrated using real-world events, social media, and visitor flow data in Hong Kong. Furthermore, a case study focusing on a specific event venue is presented to illustrate the practical implications of the research findings for informing event-aware transport policies, such as establishing special shuttle services between event venues and key border checkpoints at specific hours. The contributions of this study are summarized as follows:

- Large-scale user-generated social media content is integrated into event-aware analytical frameworks through semantic analysis facilitated by LLMs.
- Event popularity metrics are derived using social media data, considering both pre-event promotion and post-event word-of-mouth effects, which can be used to support event-aware prediction of cross-city visitor flows.
- A GBDT rolling prediction model is adapted for event-aware prediction of cross-city visitor flows under diverse and concurrent events, allowing local authorities to proactively formulate targeted measures and specialized transport policies.

- The proposed analytical framework is demonstrated using real-world events, social media, and visitor flow data from Hong Kong, the results reveal heterogeneous effects across different event types and major travel modes, with further discussion on potential policy implications.

2. Literature review

2.1. Public events and cross-city travel

Cross-city travel demand modeling has traditionally focused on macro-level human mobility and regular commuting demand within metropolitan areas. Researchers have employed gravity models to analyze cross-city travel patterns, considering factors such as population size, land use, and travel behaviors (Zhao et al., 2023; Wirasinghe and Kumara, 1998). Additionally, complex network analysis has been utilized to uncover the structure of cross-city commuting systems, providing insights into mobility flows and spatial interactions (De Montis et al., 2010, 2007). However, these studies primarily emphasize regular travel patterns while often overlooking scenarios where demand experiences sudden, significant fluctuations due to external disruptions.

The demand for cross-city travel often exhibits irregular fluctuations influenced by various factors, such as pandemics (Wei and Pan, 2024; Wang et al., 2022; Yamaguchi and Nakayama, 2023), extreme weather conditions (Lu et al., 2014; Yang et al., 2021), holidays (Li et al., 2020), etc. In recent years, public events, such as concerts and fireworks displays, have become an increasingly significant driver of cross-city travel demand surges. Policymakers leverage frequent and diverse events to attract visitors and boost local economy (Pike and Page, 2014). The continuous advancement of metropolitan transportation systems has further facilitated people to travel across cities more conveniently (Wang et al., 2018; Li et al., 2020). As a result, individuals are more inclined to undertake cross-city trips to participate in various events. Some popular events unexpectedly attract large crowds, placing significant strain on cross-city transportation infrastructure and public transit systems. Given the growing role of events as a motivator for cross-city travel, their impact should be integrated into the analysis of cross-city visitor flows.

2.2. Impact of public events on travel demand

Previous research on the impact of public events on travel demand has largely concentrated on events occurring at a single venue. These studies aim to decompose the influence of such events and predict travel demand during event periods (Pereira et al., 2015; Rodrigues et al., 2018; Chen et al., 2020; Liang et al., 2024). For instance, an ARIMA model has been applied to predict metro passenger flows surrounding the Nanjing Olympic Sports Center over several event days (Chen et al., 2020). To evaluate the effect of events around a large venue, researchers decomposed the impact of these events alongside regular commuting patterns, and their models were tested using metro ridership data in Singapore (Pereira et al., 2015; Rodrigues et al., 2018).

In the field of tourism demand forecasting, the concept of event-motivated cross-city travel dates back to the nineteenth century (Pike and Page, 2014). However, fewer studies have examined public events as a factor that disrupts visitor flows in demand predictions (Heller and Stephenson, 2021; Zhang et al., 2024). For example, Zhang et al. (2024) integrated disturbances from both crises and mega-events into tourism demand forecasting for two cities, using online search trends related to events as predictive variables. Nevertheless, this study was limited to examining four infrequent international mega-events. Today, policymakers have increasingly relied on more frequent and diverse events to attract visitors and maintain urban vibrancy, such as music concerts and exhibitions. The joint impact of multiple concurrent events on city-wide visitor flows remains a largely unexplored area of research.

2.3. Opportunities and challenges in integrating social media data

Events can vary significantly in scale, from large music concerts to small exhibitions. Estimating attendance is particularly challenging for events that are held in open spaces or do not require reservations. To quantify an event's potential to attract crowds and incorporate it as a measurable feature, prior studies employed topic modeling techniques to categorize event topics or used more recent approaches, such as LLMs, to interpret events' potential popularity based on textual event descriptions (Rodrigues et al., 2018; Liang et al., 2024). However, social media engagement data, such as the number of "likes" of relevant posts, provides additional information on an event's online popularity and can be integrated with its textual descriptions to estimate the event's potential to attract crowds (Rodrigues et al., 2018; Liang et al., 2024; Pereira et al., 2015; Zhang et al., 2024).

In recent years, social media platforms such as Facebook and Instagram have become essential marketing channels for event organizers and marketing organizations (Zhou, 2024). Events are typically one-off or periodic, resulting in fewer information sources available for potential visitors (Hu et al., 2022). When planning trips motivated by events, users often turn to these social media channels for event-related information. Social media data encompasses not only event advertisements from event marketing organizations but, more importantly, user-generated content that shares participants' personal experiences. Studies indicate that social media information affects user behavior through two mechanisms: the attention effect and the endorsement effect (Hu et al., 2022; Cui et al., 2018; Duan et al., 2008; Dellarocas et al., 2007). Promotional posts, created by event organizers or marketers, enhance the events' online exposure, contributing to the attention effect. User-generated content, which shares attendees' personal experiences, serves as an endorsement and fosters online word-of-mouth communication. Driven by both online promotional efforts and extensive

Table 1
Incorporation of social media data in studies on event-related mobility.

Authors	Topics	Data Sources	Relevance Identification	Feature Extraction
Ni et al. (2016)	Real-time traffic event detection and passenger flow predictions within the following hour	Twitter	Geo-tags, time window, hashtags	Number of event-related posts and unique tweet users
Xue et al. (2022)	Passenger flow predictions under events within the following hour	Sina Weibo	Geo-tags, time window	Number of event posts
Pereira et al. (2015)	Event-related crowding explanation	Facebook, Google	Search through geo-tags and hashtags, and filter manually	Event topics, categories, Facebook “likes”, Google hits
Du et al. (2014)	Event Attendance prediction in Event-Based Social Networks	Douban	–	User interest with a forgetting function

word-of-mouth communication, these posts jointly impact people’s intentions to travel across cities to attend events (Boivin and Tanguay, 2019; Hudson et al., 2015; Zeng and Gerritsen, 2014; Xiang and Gretzel, 2010; Dedeoğlu et al., 2020).

However, social media data are noisy and often involve irrelevant posts. Table 1 summarizes studies that integrate social media information to predict travel demand during events and outlines their methods for identifying event-related posts. For short-term prediction of travel demand (e.g., next-hour prediction), researchers usually identify event-related posts based on spatiotemporal proximity to capture real-time attendance behaviors (Ni et al., 2016; Xue et al., 2022). These studies primarily focus on social media check-in records that disclose users’ real-time locations, reflecting their event-going behaviors. Specifically, this approach first selects posts generated a few hours before the event start time around the event venue, and then incorporates features such as the number of posts and unique post creators into prediction models (Ni et al., 2016; Xue et al., 2022). However, it relies mainly on the spatiotemporal attributes of social media posts, with less emphasis on the semantic relevance of the posts.

In contrast, to effectively plan and manage the sudden surge of travel demand due to events, it is necessary to predict visitor flows at least a few days in advance. In this case, the focus shifts from social media check-ins to posts that promote the event or share personal experiences, as they influence other people’s intentions to attend. Since these promotional and experience-sharing posts are not necessarily posted at the event venue or within the event’s time frame, it is no longer sensible to filter social media posts based on spatiotemporal relevance. Instead, semantic relevance becomes more important, which requires new methods to comprehend the content of social media posts rather than their time or location. Although previous studies have used variables such as Facebook “likes” and Google hits to explain metro ridership during events (Pereira et al., 2015), they still largely relied on manual filtering and only considered posts from marketing organizations’ official event pages or accounts, overlooking the significant influence of user-generated content. This is problematic as people’s perceptions of events are shaped not only by official marketing but also by word-of-mouth communication on social media (Hudson et al., 2015; Lei et al., 2023). The reliance on manual screening further limits the method’s scalability. Therefore, there is a pressing need for more efficient and reliable data processing methods to enhance semantic analysis for large-scale user-generated content.

2.4. Advantages of employing LLMs in social media data analysis

In recent years, the field of natural language processing has experienced a significant transformation with the emergence of LLMs, such as the Generative Pre-trained Transformer (GPT) series (OpenAI, 2023). LLMs demonstrate exceptional abilities in understanding and generating human-like text. In relevant fields, Liang et al. (2024) highlighted the proficiency of LLMs in summarizing event descriptions, demonstrating their performance to be comparable to humans for this specific task. In addition, LLMs have been utilized in tasks such as understanding urban concepts (Fu et al., 2024) and identifying location descriptions (Mai et al., 2024).

LLMs have the potential to significantly enhance the efficiency and applicability of large-scale social media data analysis when compared to conventional Natural Language Processing (NLP) methods. A major challenge in identifying relevant social media posts from noisy raw data is the disparity in linguistic style and register between event information and user-generated content. Event information often employs formal language, whereas user-generated content is frequently informal and highly variable in expression (Eisenstein, 2013). LLMs are capable of assessing semantic similarity or relevance across different textual data sources with substantially diverging linguistic styles. Traditional NLP approaches, which rely primarily on surface-level similarities, such as lexicon-based methods (Jiang and Conrath, 1997; Mihalcea et al., 2006), often fail to recognize relevance when their descriptions differ at the lexical level but are semantically aligned.

Even advanced contextual embedding methods based on pre-trained language models, such as Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al., 2019) and Sentence-BERT (Reimers and Gurevych, 2019) have substantially improved contextual understanding. However, their ability to generalize across documents remains limited. Typically, they require fine-tuning on task-specific data to effectively capture context. However, the quality and breadth of their understanding are heavily limited by the scope and diversity of available fine-tuning datasets. In contrast, LLMs exhibit substantially greater generalizability, with strong semantic comprehension and analytical abilities even in zero-shot or few-shot settings, without the need for extensive fine-tuning (Brown et al., 2020). This makes LLM-based embeddings particularly advantageous for scenarios involving heterogeneous text and

diverse phraseology, as in this study. Identifying semantic relevance between event descriptions and social media content could enhance the quality of social media data and the performance of visitor flow prediction (Song and Zhang, 2024).

2.5. Summary of research gaps

To summarize, this study aims to address two major research gaps. First, while public events are commonly recognized as a major source of irregular visitor flows and sudden travel demand surges, previous research has primarily focused on the impact of occasional international mega-events (e.g., Olympic Games), or the effects of events at a specific event venue (e.g., a major sports stadium). The joint effects of diverse and concurrent events on cross-city visitor flows remain largely unexplored in the existing literature. Second, social media data provides a valuable resource for assessing the potential popularity of events. However, large data volume and inherent data noise present significant challenges for systematic analysis. Previous studies have relied on manual filtering of relevant social media posts, which limits the scalability of semantic analysis for large-scale applications. More efficient data mining and feature extraction methods are needed to leverage social media data for estimating events' potential to attract visitors.

To address the above research gaps, the rest of this paper will present an analytical framework that (1) utilizes LLMs to efficiently match social media posts to each individual event and derive event-specific popularity scores based on relevant social media engagement metrics (for the second research gap), and (2) incorporate these popularity scores in dynamic cross-city visitor flow prediction under diverse and concurrent events (for the first research gap).

3. Study area and datasets

3.1. Study area

We choose Hong Kong as the study area due to its strategic policy focus on hosting diverse public events to attract visitors and drive economic growth. Located in southern China within the Greater Bay Area (GBA), Hong Kong maintains close geographical ties and experiences frequent flows of people and economic activities with Chinese Mainland. As shown in Fig. 1, Hong Kong currently operates four major land customs entry points and two metro entry points along the border with Shenzhen, one of the biggest cities in China. Additionally, a cross-sea bridge checkpoint connects Hong Kong with Macau and Zhuhai; four ferry entry points link cities in the region. West Kowloon High-Speed Rail (HSR) Station and Hong Kong International Airport provide direct links to numerous Chinese Mainland cities. These transportation hubs serve as primary channels for cross-border visitors from Chinese Mainland.

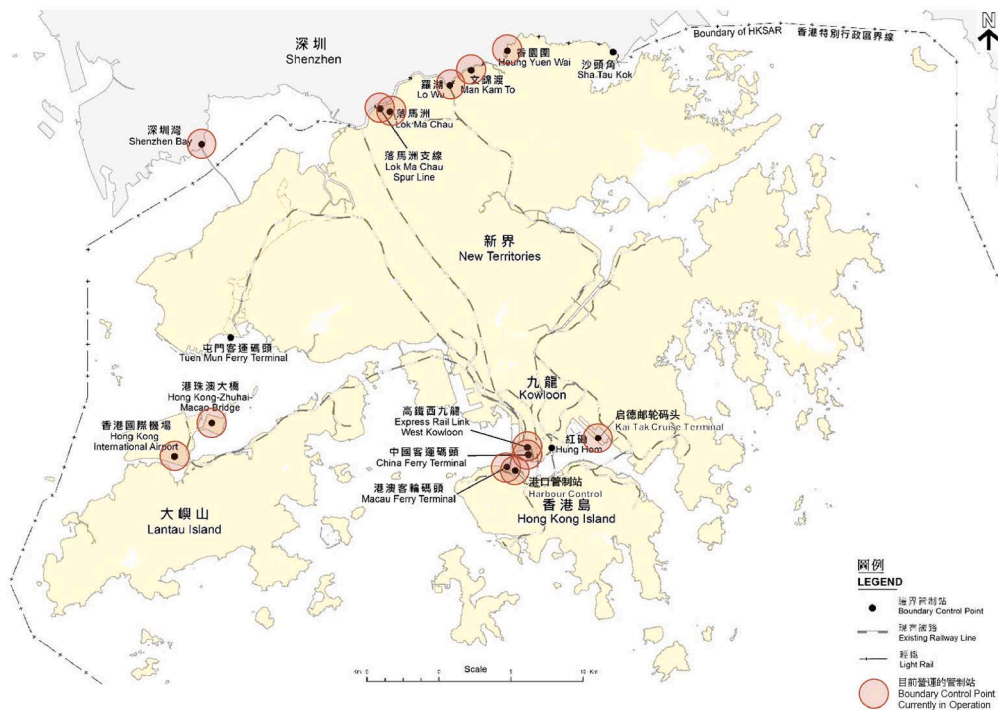


Fig. 1. Cross-border entry points in Hong Kong (adapted from Hong Kong Planning Department, 2021).

Hong Kong has been a popular destination for visitors from Chinese Mainland, especially those coming from neighboring cities. In 2017, Chinese Mainland visitors contributed to an average daily arrival rate of 216,600 in Hong Kong, with 57.8% being same-day trips and 91.8% of these visitors originating from GBA (excluding Hong Kong residents) (Hong Kong Planning Department, 2017).

However, the outbreak of the COVID-19 pandemic severely disrupted leisure travel by Chinese Mainland visitors ([Hong Kong Planning Department, 2017, 2021](#); [Tsui et al., 2021](#)). Following the reopening of customs in January 2023, Hong Kong has strategically hosted events to attract visitors and increase tourism revenue ([Getz and Page, 2016](#); [Chang et al., 2022](#)). These events have attracted substantial numbers of Chinese Mainland visitors, putting pressure on customs processing and accommodation capacities. As transportation services are typically designed for regular commuting traffic, the sudden surge of visitor arrivals can lead to overcrowding and travel delays, negatively affecting the visitors' experience. Predicting visitor flows for upcoming events is thus essential for managing visitor demand, as it allows for proactive adjustments, such as assigning additional frontline staff at key entry points, enhancing local public transit service capacities, and deploying traffic management measures around event sites.

3.2. Datasets

3.2.1. Visitor flows, holiday and weather data

Since cross-border arrivals from Chinese Mainland resumed in March 2023, we collected data from March 1, 2023, to May 15, 2024. The cross-border visitor arrival data is provided by the Hong Kong Immigration Department.¹ Fig. 2 illustrates daily cross-border Chinese Mainland visitor arrivals from January 2023 to May 2024, showing the resurgence since the borders reopened. Visitor arrivals exhibit noticeable weekly peaks and sharp increases during calendar events (labeled in red). Additionally, January–February and July–August are usually the summer and winter school vacation periods in China, respectively. During these periods, visitor arrivals on weekdays experience substantial increases. Given that weather conditions are considered to significantly influence event attendance and travel behavior ([Zhang et al., 2021](#); [Becken, 2013](#)), we further obtain daily total rainfall,² maximum daily temperature,³ and historical tropical cyclone warning signals (typhoon signals) from the Hong Kong Observatory.⁴

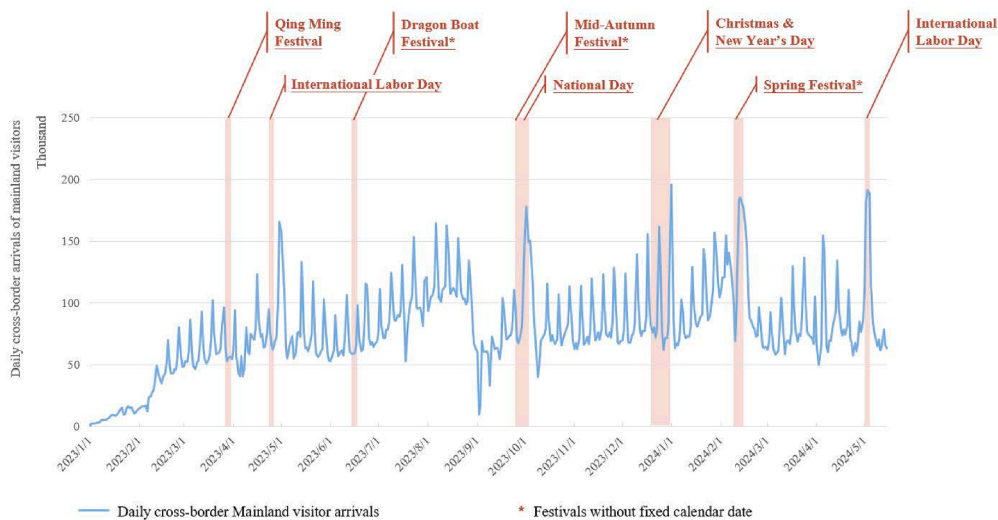


Fig. 2. Daily flows (arrivals) of Chinese Mainland visitors to Hong Kong from January 2023 to May 2024.

3.2.2. Event data

Event data is collected through multiple sources in various formats. We compile event data for Hong Kong starting from March 2023, utilizing a dedicated event website, *Timable*.⁵ This platform provides comprehensive historical event information, including concerts, festivals, exhibitions, competitions, performances, etc. The local government also displays selected events on the official tourism and destination branding websites in different structures. As a supplement, we obtain the “year-round event data” in 2023 from the *Hong Kong Tourism Board* website,⁶ the “mega events list” in 2024 provided by *Hong Kong Asia’s World City* website,⁷ and the “major sports list” from the *Major Sports Events Committee* website.⁸

Fig. 3 illustrates the event page of a fireworks display on Timable, providing details such as event title, time, location, and description. However, the raw textual event features extracted from web pages are often unstructured, which poses challenges for

¹ <https://data.gov.hk/en-data/dataset/hk-immmd-set5-statistics-daily-passenger-traffic>

² <https://data.gov.hk/en-data/dataset/hk-hko-rss-daily-total-rainfall>

³ <https://data.gov.hk/en-data/dataset/hk-hko-rss-daily-temperature-info-hko>

⁴ <https://www.hko.gov.hk/tc/wxinfo/climat/warndb/warndb1.shtml>

⁵ <https://timable.com>

⁶ <https://www.discoverhongkong.com/hk-eng/what-s-new/events.html>

⁷ <https://www.brandhk.gov.hk/en/mega-events/mega-events>

⁸ <https://www.mevents.org.hk/en/calendar.php>

Title 冬日維港水上煙火表演

Time 16-17 Dec 2023(Sat-Sun), 23-26 Dec 2023(Mon-Tue, Sat-Sun) 8:00 pm to 8:10 pm

Location 西九文化區 藝術公園 海濱長廊 維多利亞港 | 香港西九龍

Textual Description

假如你還未抵港，維港上空綻放煙火，加上迷人聖誕燈色加持，真係難得有多精彩呀！想去的現場欣賞呀，就帶把望遠鏡準備好，週末同聖誕期間，親臨下邊3大最佳觀賞地點，感受熱鬧聖誕煙火噴湧澎湃日景景！

最佳觀賞地點推薦：
九龍西九文化區藝術海濱長廊(近海怡實)
海濱城「海濱觀點」觀景台
中西區海濱長廊 - 上環段



Translation: Winter Firework Displays over Victoria Harbor

Translation:
Art Waterfront Promenade, Kowloon West Cultural District, near the sea | Hong Kong West Kowloon

Translation:
Just this past weekend, the sky over the Victoria Harbor was filled with dazzling fireworks, coupled with the mesmerizing view of the Victoria Harbor, it's not enough to take any pictures! If you want to enjoy the scene, take the opportunity to visit the following 3 best viewing spots this coming weekend and Christmas period to feel the lively Christmas fireworks and rich festive atmosphere!

Best Viewing Location Recommendations:
Art Waterfront Promenade, Kowloon West Cultural District, near the sea
Harbor City "Marine Viewpoint" Observation Deck
Central and Western Waterfront Promenade - Sheung Wan Section

Fig. 3. Illustration of the event page of winter fireworks displays on timetable.

conventional NLP models to interpret the underlying semantics and generate specific event features. For instance, many events occur over multiple sessions, and conventional models find it difficult to extract structured times for each session from expressions like “16-17 Dec 2023 (Sat-Sun), 23-26 Dec 2023 (Mon-Tue, Sat-Sun) 8:00 pm - 8:10 pm”, as shown in Fig. 3. Furthermore, event information extracted from government websites is in various formats. To process textual information from diverse sources, we employ LLMs to extract unified key features such as the event title, time, location, and description for each sub-session. The details will be described in Section 4.2.

3.2.3. Social media data

An event's popularity on social media platforms is a powerful indicator of its potential to attract visitors. Among the popular Chinese social media platforms, *Rednote* (also known as *Xiaohongshu*) has emerged as a prominent player in recent years, boasting over 350 million users. Unlike other online platforms that focus on one-way content generation for destination recommendations, such as *TripAdvisor*, or the specialized event-based social networks, such as *Douban*, *Rednote* encourages vibrant peer communication through a wealth of user-generated content. *Rednote* users typically share information or personal experiences about the latest trending events, making it a popular tool for discovering event information (Lei et al., 2023).

To assess an event's popularity, we summarize keywords for the events and search for *Rednote* posts using these keywords. Using an open-source social media crawling tool on GitHub, *MediaCrawler*,⁹ we collected the top g posts for each event, ranked by the number of “likes” received. In this case, without loss of generality, we set $g = 100$ for each event. Due to the limitations of search algorithms and keyword ambiguity, irrelevant posts are occasionally retrieved in data collection. We will introduce the LLM-enhanced method for processing social media data in Section 4.2.

As shown in Fig. 4 (a-b), each post on *Rednote* includes information such as user ID, title, textual content, hashtags, time, and counts of “like”, “collect”, and “comment”. When *Rednote* users seek information about events, they often click the “like” or “collect” buttons to express their interest and bookmark the posts. This allows them to conveniently revisit relevant information from their homepage when planning trips. We assume that the total count of “likes” and “collects” for an event's related posts reflects the public's interest in the event on social media, thereby indicating its potential to attract attendees. Based on observed user behavior patterns on this specific social media platform, the number of “comments” is less indicative of an event's online popularity, as comments are less practical for users to revisit as a future reference. Therefore, we measure an event's social media popularity by the total number of “likes” and “collects” under relevant posts on *Rednote* in this study. On most social media platforms, users typically express their

⁹ <https://nanmicoder.github.io/MediaCrawler/>

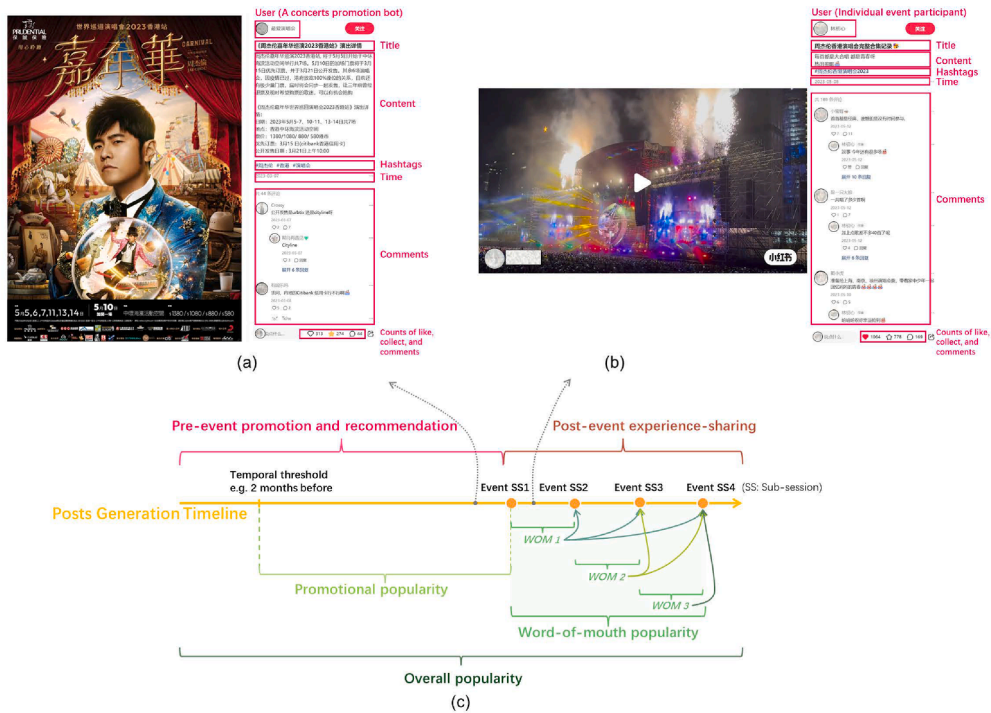


Fig. 4. Illustration of social media content and corresponding popularity measures.

interests through “likes”. Therefore, we use “likes” as a general term to refer to user engagement with social media posts. It is worth highlighting that *Rednote* is only used here as a representative social media platform, and the methods introduced in later sections should be easily adaptable to other platforms.

4. Methods

4.1. Study framework

This study proposes an event-aware analytical framework that extracts event features from online data sources and incorporates these features to predict cross-city visitor flows. The overall study framework is depicted in Fig. 5. First, we construct a comprehensive, city-specific event dataset and design a generalizable pipeline that leverages LLMs to mine features from diverse online event sources and large volumes of user-generated social media content. By integrating these event-derived features with conventional predictive variables, we adapt a GBDT rolling prediction model to forecast visitor flows for upcoming events. To identify factors influencing visitor flows, we use Shapley Additive Explanations (SHAP) values, which reveal the relative importance and impact directions of different features. Furthermore, to demonstrate how the model results can inform event-aware policymaking, we present a case study focusing on concerts at Hong Kong Coliseum, highlighting the importance of expected cross-city visitor flows for formulating targeted strategies such as special shuttle service design.

4.2. LLM-enhanced event feature mining

To effectively structure event features from various online data sources, we design a generalizable pipeline for LLM-enhanced event feature mining. Fig. 5 illustrates the detailed pipeline that integrates the advanced text processing and semantic analysis capabilities of LLMs with human expertise to tackle three main objectives: (1) filtering events, (2) matching social media posts related to events, and (3) quantifying event popularity on social media.

To filter events, we use LLMs to structure event time, summarize event descriptions, and standardize classifications across multiple data sources. Specifically, we establish heuristics to exclude events that are unlikely to draw visitors from other cities. Smaller events are typically long-term, held in smaller venues, and lack broad public appeal. The details will be described in Section 4.2.1.

To identify event-related social media posts, we characterize posts based on their titles, content, hashtags, timestamps, and the number of “likes” they receive. LLMs can be used to compare the semantics of textual data in event descriptions and social media posts, allowing us to pinpoint posts associated with specific events and filter out irrelevant content. The details will be described in Section 4.2.2.

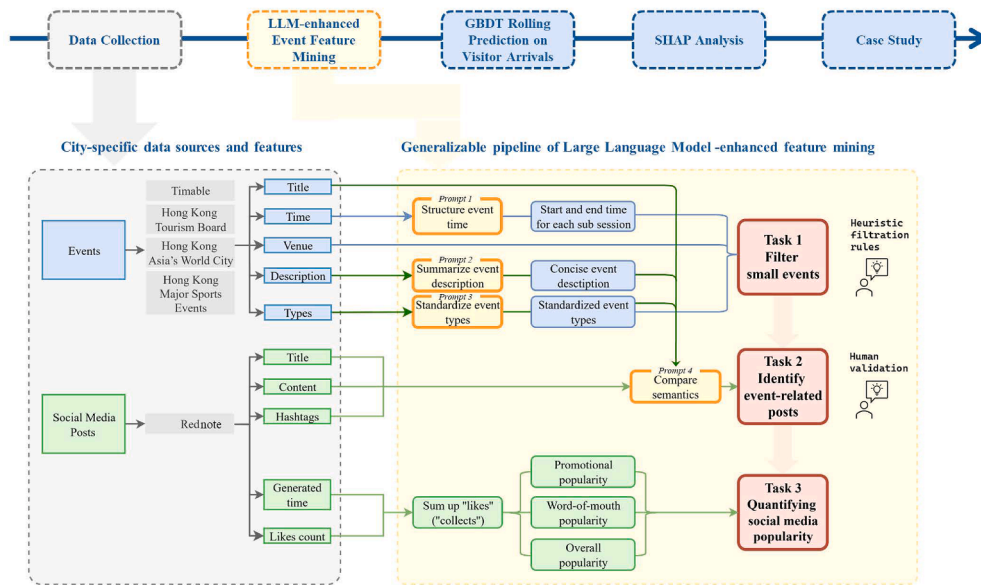


Fig. 5. Study framework with LLM-enhanced feature mining pipeline.

To measure social media popularity, we propose three metrics based on the post's temporal alignment and content intent: promotional popularity, word-of-mouth popularity, and overall popularity. Specifically, we distinguish between social media posts that promote events beforehand and those that share experiences afterward; the former are associated with promotional popularity and the latter reflect word-of-mouth popularity. Overall popularity encompasses all forms of social media engagement with the events, regardless of posting time. The effectiveness of these popularity metrics will be evaluated in [Section 4.2.3](#).

4.2.1. Feature enhancement and event filtering

Event time information is often presented as unstructured text, making it challenging to characterize. To address this, we leverage the ability of LLMs to understand semantics and convert raw event information into structured features. As shown in [Table 2](#), we design *Prompt 1* to process raw time information by extracting and formatting specific start and end times for each session. *Prompt 2* directs the LLM to summarize event descriptions while omitting details about event time, location, ticketing, and promotion. To ensure these summaries are both comprehensive and concise, we impose limits on the LLM's token usage during summary generation. These prompt templates can be adapted for specific events by modifying placeholders as needed.

Different types of events vary in their appeal to visitors, requiring consistent classification rules across various data sources. To achieve this, we predetermine the scope of event types based on existing classifications from specialized event websites and local tourism portals. As shown in *Prompt 3*, the LLM categorizes events into these predefined types, standardizing event classification.

To exclude smaller events that are less likely to attract visitors from other cities, we establish three heuristic rules. First, we limit the number of sub-sessions to exclude long-term events, such as individual art exhibitions, which typically draw local visitors continuously and are unlikely to attract a significant number of cross-city visitors in the short term. Second, we exclude events held in small venues, such as ground floors of shopping malls, by compiling a list of large-capacity venues commonly used for each event type and filtering events accordingly. Third, we focus on certain event types, including concerts, exhibitions, sports competitions, and fireworks displays, as small-scale events like weekend fairs are less likely to be promoted to visitors. These rules focus our efforts on events likely to influence visitor flows, streamlining the collection and processing of social media data.

4.2.2. Event-related social media post identification

Due to the ambiguity of search algorithms in social media platforms, irrelevant information would be captured when using keyword searches. To accurately identify social media posts related to public events, we use LLMs to compare the semantics of event titles, types, and summarized descriptions with the information found in social media posts, including titles, content, geo-tags, and hashtags. As illustrated in [Table 2](#), *Prompt 4* assists in determining whether a post pertains to an event taking place in the city within a specific time frame.

To assess the reliability of the LLM-based semantic analysis method, we use a stratified sampling approach to select 5% of events from each event category, along with their corresponding social media posts. The performance of LLMs in assessing the relevance of posts to a specific event is compared with both the manual screening method as used in prior studies and a conventional NLP approach, Sentence-BERT. The LLM is found to achieve over 93% agreement with human evaluators, outperforming Sentence-BERT. Further details are provided in [Appendix A](#).

Table 2
LLM prompt templates for event feature mining

ID	Task	Prompt template
Prompt 1	Structuring event period	Below is the raw description of an event's hosting time, including event ID: <i>{event ID}</i>, and event time: <i>{raw time description}</i>. Your task is to extract the exact start and end datetime for each session. If an event has multiple sessions, find the exact start and end datetime for each session. If a session does not have a specific start and end time, set the start time to "00:00:00" and the end time to "23:59:59" of the event day. Provide the output in four columns: • Id: The event ID. • Sub id: A unique number for each session of the event, starting from 1. • Start time: The start datetime of the event session in "YYYY-MM-DD hh:mm:ss" format. • End time: The end datetime of the event session in "YYYY-MM-DD hh:mm:ss" format.
Prompt 2	Summarizing event descriptions	Please summarize the event information given the raw event description extracted from the web page: <i>{raw event description}</i>. Requirements are as follows: • Exclude details about time, location, registration, tickets, and payment; only focus on the event content. • Be concise within <i>{token usage}</i> tokens. • Summarize in <i>{language}</i>. • Only provide the summarized event description without any additional explanations.
Prompt 3	Standardizing event classification	Below is a description of an event that occurred in <i>{study area}</i>, called <i>{event title}</i>. <i>{summarized event description}</i>. Please match the events into the following categories: <i>{A list of predefined event types}</i>. Only provide the event type without any additional description.
Prompt 4	Identifying event-related social media posts	Below, you will find a description of an event and the content of a social media post. Please compare their semantics and determine if the post post ID is related to the given event. Respond with "Yes" if the post is related, or "No" if it is not. Event Information: • Title: <i>{event title}</i> • Type: <i>{event type}</i> • Description: <i>{summarized event description}</i> Social Media Post Details: • Title: <i>{post title}</i> • Content: <i>{post content}</i> • Geo-tags: <i>{post geo-tags}</i> • Hashtags: <i>{post hashtags}</i> Criteria for Relevance: • The post must reference an event occurring in <i>{study area}</i>. • The post's content must demonstrate a clear connection to the event based on the provided event information.

To test the robustness of our LLM-based method against minor prompt variations, we design alternative prompts with different phrasing and levels of detail, assessing performance based on agreement rates with human evaluators. It is found that all prompts yield more than 88% alignment with human validation, with stepwise and more detailed instructions performing slightly better. Detailed results are presented in [Appendix B](#).

4.2.3. Social media popularity

To measure an event's popularity on social media, we distinguish between overall, promotional, and word-of-mouth popularity. Overall popularity is defined as the aggregate number of "likes" across all event-related posts, regardless of posting time. We further classify posts into two categories based on their chronological relationship to the event: pre-event promotional posts and post-event experience-sharing posts. To ensure the posts are pertinent to the current event, we establish a temporal threshold to exclude posts from previous years or seasons. The "likes" on pre-event promotional posts indicate an event's popularity before it happens and reflect the effectiveness of promotion efforts, which we term promotional popularity.

After attending an event, attendees frequently share their experiences on social media. This creates a word-of-mouth effect that influences the attendance of others in future sessions. As illustrated in [Fig. 4 \(c\)](#), $WOM_{n'}$ represents the aggregate "likes" on social media posts shared between session n' and the next. These experience-sharing posts are recommended to potential attendees, who

may choose to attend any of the remaining $N - n'$ sessions, where N denotes the total number of sessions in the event. Based on this mechanism, we define word-of-mouth popularity ($WOMP_n$) for session n as the cumulative impact of all preceding sessions:

$$WOMP_n = \sum_{n'=1}^{n-1} (\alpha_{n,n'} \cdot \frac{WOM_{n'}}{N - n'}) \quad (1)$$

where $n \in \{2, \dots, N\}$ denotes the current session number. Since the word-of-mouth effect is not applicable to the first session or single-session events, we calculate $WOMP_n$ starting from $n = 2$. In this framework, $\alpha_{n,n'}$ serves as a weight factor that models the temporal dynamics of the word-of-mouth impact. Drawing on previous research suggesting that the effect of online word-of-mouth tends to diminish over time (Delre and Luffarelli, 2023), we propose and compare three decay patterns for $\alpha_{n,n'}$: uniform influence, exponential decay, and linear decay. Among these, the exponential decay pattern yields the best model performance. In this case, the weight factor is defined as:

$$\alpha_{n,n'} = e^{-\beta(n-n')} \quad (2)$$

where the parameter β controls the rate of decay and is fixed at 0.2. Further details of tested temporal dynamic patterns and decay rates are provided in Appendix C.

To evaluate the effects of promotional popularity and word-of-mouth popularity on visitor flow prediction, we will test multiple model specifications, as detailed in Table 3.

4.2.4. Parameter setting

The parameter settings in the LLM-enhanced event feature extraction process can be adjusted according to the requirements of practical applications. The parameter setting for this study is specified below:

- The LLM, {GPT-4}, is used to enhance event filtration and feature extraction processes (OpenAI, 2023). Our study focuses on events held in {Hong Kong} during {the years of 2023 and 2024}.
- For summarizing event descriptions in Prompt 2, we limit the token usage to {120} per event. Following the original language used in online sources, the summaries are produced in {Chinese}.
- In Prompt 3, we define a list of event types based on the existing classifications from dedicated event websites and local tourism websites: Timable, Hong Kong Tourism Board, Hong Kong Asia's World City, and Hong Kong Major Sports Events. The predefined event types include {music concerts, exhibitions, sports competitions, fireworks displays, fairs, performances, and religious activities}.
- In filtering small events, we keep the event types in {music concerts, exhibitions, sports competitions, and fireworks displays}. The maximum number of sub-sessions is set to {30}, assuming events with more than 30 sub-sessions are less likely to cause significant travel demand surges.
- To quantify social media popularity, we apply a temporal threshold, filtering out posts created more than {two months} before the event.

4.3. Key variables

To understand the factors influencing visitor flows, our model includes variables such as holiday features, weather conditions, and temporal trends. To explore how different types of events attract visitors, we also include the event's social media popularity features by type. The explanatory variables are summarized in Table 3.

4.3.1. Holiday features

Visitor flows exhibit clear weekly cyclical patterns, along with significant peaks during major holidays. To capture the day-of-week effect, we include six binary variables to indicate each day of the week, excluding Tuesday (omitted as a reference category). Holiday effects are determined based on the primary origin of visitors, which, in this context, is Chinese Mainland. The major holidays considered in the study include the Spring Festival, Qingming Festival, International Labor Day, Dragon Boat Festival, Mid-autumn Festival, National Day, Christmas, and New Year's Day. Typically, longer holidays attract more visitors, with more visitor flows at the beginning of the holiday period compared to the end. To incorporate the influence of holiday length, we calculate the number of days remaining until the holiday concludes (e.g., a value of 3 signifies three days remaining, including the present day, whereas 0 denotes a standard weekday). Additionally, major holidays can cause fluctuations outside the usual patterns, prompting the inclusion of two binary variables: one indicating the day before a holiday, and another identifying the week surrounding a holiday, either before it starts or after it ends. We also account for the proximity to holidays by including a variable that tracks the number of days to the nearest holiday, whether previous or upcoming.

Peak tourism seasons often align with school holidays, which we capture using a binary variable labeled school holidays. This reflects special periods in the academic calendar of primary and secondary schools. Specifically for the GBA areas during the study period, the summer vacation spans from July 9, 2023, to August 31, 2023, and the winter vacation stretches from January 22, 2024, to February 19, 2024. Including these variables allows the model to accommodate the impact of both regular and seasonal variations in visitor flows.

4.3.2. Weather features

Weather conditions significantly influence travel behavior. We include variables for daily rainfall measured in millimeters, daily maximum temperature measured in Celsius, and typhoon days when a tropical cyclone warning signal is issued. In total, three variables are included in the analysis.

4.3.3. Temporal trends

Temporal trends provide crucial indicators of seasonal fluctuations in visitor flows. To capture these trends, especially when historical data is limited, we introduce the Weighted Moving Average (WMA) changing rate as a feature. The WMA smooths the time series data by assigning progressively smaller weights to older observations, thereby emphasizing recent data points. This approach helps mitigate the impact of short-term fluctuations and highlights the underlying trend.

The WMA M_t for visitor flow on day t over a window of P days is calculated as:

$$M_t = \frac{\sum_{p=0}^{P-1} \mu_p \cdot x_{t-p}}{\sum_{p=0}^{P-1} \mu_p} \quad (3)$$

where x_{t-p} represents the visitor flow on day $t - p$, and μ_p denotes the weight assigned to that day. In this study, we apply a linear decreasing weight $\mu_p = P - p$.

To predict visitor flows for day t , we utilize the changing rate of the WMA from day $t - 1$ to capture the most recent trend. This rate, C_t , is defined as:

$$C_t = \frac{M_{t-1} - M_{t-2}}{M_{t-2}} \quad (4)$$

By incorporating the WMA changing rate over the past $P = 10$ days, the model captures the momentum of visitor flow trends, providing critical information regarding the direction and strength of recent movements.

4.3.4. Event features

As discussed in Section 4.2.3, we establish three measures to assess event popularity on social media platforms: overall popularity, promotional popularity, and word-of-mouth popularity. We calculate these measures by event type, including music concerts, exhibitions, fireworks displays, and sports competitions. Notably, fireworks displays in this dataset are mostly single-session events held during major festivals in Hong Kong. Since the word-of-mouth effect does not apply to new or single-session events, we exclude word-of-mouth popularity for fireworks displays to adapt to this dataset.

To thoroughly evaluate the effectiveness of event popularity measures, we compare model performance across five distinct feature sets. As outlined in Table 3, *Feature Set 1* includes holidays, weather conditions, and temporal trends, while deliberately excluding event-related features. Building upon *Feature Set 1*, *Feature Set 2* incorporates the count of events by type, based on the assumption that the number of events held on a particular day could intuitively correlate with visitor numbers. To assess the impact of different popularity features, *Feature Set 3* extends *Feature Set 1* by adding the overall popularity of events categorized by type, *Feature Set 4* extends *Feature Set 1* by including promotional popularity of events. Finally, *Feature Set 5* integrates both word-of-mouth and promotional popularity features. Through these diverse feature sets, we aim to examine the comparative effectiveness of event popularity measures in predicting visitor flows.

4.4. Visitor flows prediction model

We develop a rolling prediction model based on GBDT to predict visitor flows. By incorporating a rolling window approach with a customized sample weighting mechanism, our model maintains adaptability and accuracy in the face of irregular demand dynamics during events (Liu et al., 2023). Originally popularized by Friedman (2001), GBDT constructs an ensemble of decision trees in a sequential fashion, with each new tree aimed at correcting the prediction errors of the existing ensemble. This method excels at capturing non-linear relationships and feature interactions, making it suitable for managing data characterized by irregular distributions and sparsity. These data characteristics are often present in event features, where time series values are frequently zero or missing, particularly concerning event popularity variables that exhibit sparsity and a long-tailed distribution. Thus, GBDT is particularly well-suited for capturing the effects of these features on visitor flows and providing robust predictions. Model details are provided in Appendix D.1.

To manage the irregular dynamics observed in visitor flows, we employ a rolling prediction method. This approach involves iterative training and testing in a rolling window, which mirrors real-world scenarios where operators continuously collect new data and predict visitor flows for subsequent days. Given a prediction horizon φ , the process begins by defining the training data from the initial time step 1 to time t . We then predict values for the period extending from $t + 1$ to $t + \varphi$. As the rolling prediction window slides by each time unit, the process is repeated, allowing the model to adapt to the latest data points. For every time step, the model generates φ predictions, subsequently averaging these predictions to provide a robust prediction. This approach accommodates the fluctuating dynamics of visitor flows as new information becomes available. To assess the robustness of this method in real-world operations, we compare rolling prediction horizons φ ranging from 1 to 7 days. Long prediction horizons allow transportation authorities to gain sufficient time to collaborate closely with cross-city infrastructure operators, security personnel, emergency sectors, and event organizers to implement well-coordinated management plans.

Table 3
Description of key variables.

Category	Features	Description	Feature Set 1	Feature Set 2	Feature Set 3	Feature Set 4	Feature Set 5
Holiday features	Holidays remaining	•The number of holidays remaining according to the public holiday calendar of the tourists' origin area (e.g., 3 means three holidays remain including today, and 0 represents a weekday).	✓	✓	✓	✓	✓
	Day before a holiday	•A binary variable representing the day before a long public holiday (excluding weekends).	✓	✓	✓	✓	✓
	Week near a holiday	•A binary variable representing the week before and after a long public holiday (excluding weekends).	✓	✓	✓	✓	✓
	Days to the nearest holiday	•The number of working days to the last or next holiday (excluding weekends).	✓	✓	✓	✓	✓
	School holidays	•A binary variable indicating the summer and winter vacations of most primary and middle schools in GBA during the study period.	✓	✓	✓	✓	✓
Weather	Rainfall	•The amount of daily total rainfall measured in millimeters.	✓	✓	✓	✓	✓
	Maximum temperature	•The daily maximum temperature measured in Celsius.	✓	✓	✓	✓	✓
	Typhoon	•A binary variable representing days when a tropical cyclone warning signal is issued in Hong Kong.	✓	✓	✓	✓	✓
Temporal trend	Changing rate	•The changing rate of the Moving Weighted Average of visitor flows over the past 10 days.	✓	✓	✓	✓	✓
Event features	Event count	•The numbers of music concerts, fireworks displays, exhibitions, and sports competitions		✓			
	Overall popularity	•The overall popularity of music concerts, fireworks displays, exhibitions, and sports competitions			✓		
	Promotional popularity	•The promotional popularity of music concerts, fireworks displays, exhibitions, and sports competitions				✓	✓
	Word-of-mouth popularity	•The word-of-mouth popularity of music concerts, exhibitions, and sports competitions (fireworks displays are not included due to the scarcity of multi-session events).					✓

Considering the tourism rebound in the post-COVID era, recent trends serve as more reliable indicators of future visitor flows. To leverage this, we incorporate a sample weighting mechanism into our predictive model. Specifically, we define a sample weight function that assigns greater weights to more recent data. When comparing alternative decay patterns, the linear decay function with a decay rate of $\delta = 0.005$ demonstrates the best performance among all tested forms. The sample weight w_t for a given day t is determined by Eq. (5):

$$w_t = \max(1 - (T - t)\delta, 0) \quad (5)$$

where T is the total length of training data. Details of the comparative analysis are provided in Appendix D.2.

To evaluate the performance of rolling predictions, we calculate the Mean Absolute Error (MAE) and the R-squared (R^2) metrics based on the average prediction for each iteration over the prediction horizon φ . Let $\hat{y}_t^{\text{avg}}$ represent the average prediction for time step t over all iterations:

$$\hat{y}_t^{\text{avg}} = \frac{1}{\varphi} \sum_{k=t-\varphi}^t \hat{y}_k \quad (6)$$

where $\hat{y}_k$ represents the prediction for time step k .

The MAE is calculated as:

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t^{\text{avg}}| \quad (7)$$

where y_t represents the actual values at day t .

The R^2 score is defined by:

$$R^2 = 1 - \frac{\sum_{t=1}^T (y_t - \hat{y}_t^{\text{avg}})^2}{\sum_{t=1}^T (y_t - \bar{y})^2} \quad (8)$$

where $\bar{y}$ is the mean of the actual values across all observations.

To evaluate the effectiveness of the GBDT rolling prediction model, we compare it against several commonly used prediction models as baselines: eXtreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), Light Gradient Boosting Machine (LightGBM), Random Forest (RF), Linear Regression (LR), and AutoRegressive Integrated Moving Average (ARIMA). To ensure a fair comparison, rolling prediction and sample weighting mechanisms are applied to the baseline models (not applicable to ARIMA). Additionally, we conduct comprehensive grid searches to identify the optimal hyperparameters for each model. Further details regarding these baseline models are provided in [Appendix D.3](#).

5. Results

5.1. Data characteristics

After event filtration, we compile an event dataset comprising 368 events and 1112 sub-sessions, spanning over 14 months from March 1, 2023, to May 15, 2024. [Fig. 6](#) presents the distribution of events throughout the study period, with event types differentiated by color.

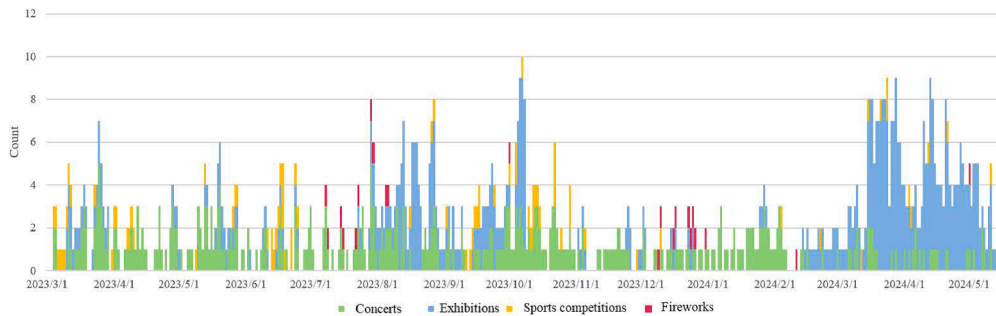


Fig. 6. Distribution of events during the study period.

[Table 4](#) presents the characteristics of collected social media posts related to these events. During the study period, concerts account for the largest number of events and generate the highest volume of social media posts. Among individual event types, posts about single occurrences of fireworks events are the most numerous, and these posts are also associated with the highest level of popularity. For concerts, fireworks, and sports competitions, the number of posts generated after the first session of the event is significantly higher than the number of promotional posts created prior to the event. It is observed that some post-event online discussions about sports competitions focus on the outcomes of the games rather than the experience of attending the event. In contrast, exhibition-related posts exhibit lower online popularity compared to posts for other events, with similar levels of engagement both before and after the event.

Table 4
Data summary statistics.

Event types	Event count	Number of social media posts	Number of social media posts per event	Average number of pre-event posts	Average number of after-event posts	Average pre-events popularity	Average after-events popularity
Concerts	216	2500	12	5	7	846	3114
Fireworks	6	632	105	32	73	16,890	37,754
Exhibitions	91	849	9	3	6	272	320
Sports competitions	55	303	6	2	4	309	2133

Note that “after-events posts” refer to those posts created after the first session of the event.

5.2. Feature sets assessment and model comparison

To predict cross-border visitor flows from Chinese Mainland, we employ a rolling prediction model utilizing all data available up to each predictive step. We conduct predictions using seven commonly used models across various feature sets over a one-day horizon. Table 5 details the performance of each model and feature set combination. Notably, the GBDT model with *Feature Set 5* achieves the highest accuracy, with an R^2 of 85.7%.

Incorporating promotional popularity and word-of-mouth popularity features, *Feature Set 5* enhances the R^2 by 3% compared to the same model without event features (*Feature Set 1*). While *Feature Set 2* (with event counts) provides some improvement compared to *Feature Set 1*, it falls short compared to other sets with event popularity features. *Feature Set 3* does not distinguish the posting time of social media content, relying purely on overall popularity, whereas *Feature Set 4* focuses only on promotional popularity, excluding word-of-mouth effects. The predictive performance of *Feature Sets 3 and 4* is slightly inferior to *Feature Set 5*, underlining the advantages of accounting for both promotional and word-of-mouth effects on social media for accurate visitor flows prediction.

For the XGBoost, CatBoost, and LightGBM models, their best predictive accuracy is also achieved using *Feature Set 5*, with nearly identical performance (an R^2 of 84.6%). The relatively superior performance of GBDT may be attributed to the algorithm's suitability for small sample sizes. Note that data sparsity is often an issue constraining empirical analysis of real-world public events. In such cases, GBDT employs first-order optimization, which typically yields smoother and more stable convergence. Its incremental learning process generates tree structures that are less sensitive to noise and exhibit higher stability. The RF model, which reduces variance through random sampling and model averaging, demonstrates limited ability to capture complex non-linear patterns and thus exhibits only moderate performance, with R^2 values exceeding 80%. In contrast, the LR and ARIMA models perform best with *Feature Set 2*. This discrepancy is due to the LR and ARIMA models' limitations in effectively processing the sparsely and irregularly distributed event popularity variables.

Table 5
Model results with different feature sets.

Feature Combinations		GBDT	XGBoost	CatBoost	LightGBM	RF	LR	ARIMA
<i>Feature Set 1</i> : Without event features	R^2 (%)	82.7	80.5	81.6	79.6	79.9	62.7	67.0
	MAE	8970	9797	9137	9776	9651	12,292	11,102
<i>Feature Set 2</i> : With event count	R^2 (%)	84.6	83.0	81.6	83.1	80.3	64.9	70.8
	MAE	8870	9463	9487	9460	9685	12,216	10,745
<i>Feature Set 3</i> : With overall popularity	R^2 (%)	85.2	83.0	82.7	83.9	80.6	22.2	36.6
	MAE	8560	9444	9216	9166	9579	13,166	12,143
<i>Feature Set 4</i> : With promotional popularity	R^2 (%)	85.2	83.7	83.0	83.6	80.6	57.7	68.5
	MAE	8714	9252	9106	9239	9643	12,613	11,232
<i>Feature Set 5</i> : With promotional and word-of-mouth popularity	R^2 (%)	85.7	84.6	84.6	84.6	80.9	56.4	67.8
	MAE	8623	9038	8722	8940	9533	12,986	11,533

5.3. Event-aware visitor flows prediction

To test the model's robustness in real-world applications, we predict cross-border visitor flows over horizons ranging from 1 to 7 days, which emulates how operators gradually incorporate newly available data points in prediction and make arrangements proactively. While longer horizons provide more time for operational arrangements, they typically yield lower accuracy. Fig. 7 illustrates the performance of the models across different prediction horizons with *Feature Set 5*. The GBDT model, represented in red, outperforms the other models, exhibiting the lowest MAE (8623 ~ 10,427) and the highest R^2 scores (85.7% ~ 78.9%). CatBoost exhibits performance close to GBDT, followed by XGBoost, LightGBM, and Random Forest. The ARIMA and Linear Regression models significantly underperform compared to the ensemble learning methods. Fig. 8 illustrates the predictions generated by GBDT model over a one-day horizon, where the predicted visitor flows are depicted in red, and the actual visitor flow is represented in blue. Although predicting irregular travel demand surges is always challenging without sufficient historical data, our event-aware model achieves a reasonable accuracy.

5.4. Factors affecting visitor flows

Understanding the factors influencing visitor flows is crucial for developing effective operational strategies for intercity transport infrastructure and services. We employ SHAP values to explain the importance and effects of the different features for the best model (GBDT under a one-day horizon with *Feature Set 5*). SHAP values interpret each feature's impact on cross-border visitor flows. In Fig. 9, the y-axis ranks features by descending importance, and the x-axis shows their impact on visitor flows. Features with clusters extending far from the center line have stronger impacts, with positive impacts to the right and negative impacts to the left.

Among the features, the number of days remaining in the holiday is the primary driving factor affecting cross-border visitor flows, with a large number of visitors arriving at the beginning of a holiday. The WMA changing rate, which indicates recent trends, also plays a crucial role in visitor flow prediction. School holidays, such as summer and winter breaks, typically correspond to the peak

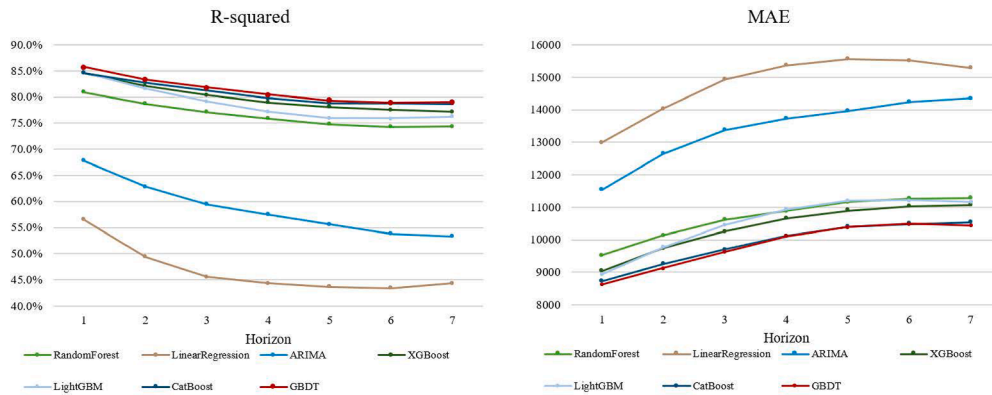


Fig. 7. Model comparison of prediction accuracy under different horizons.

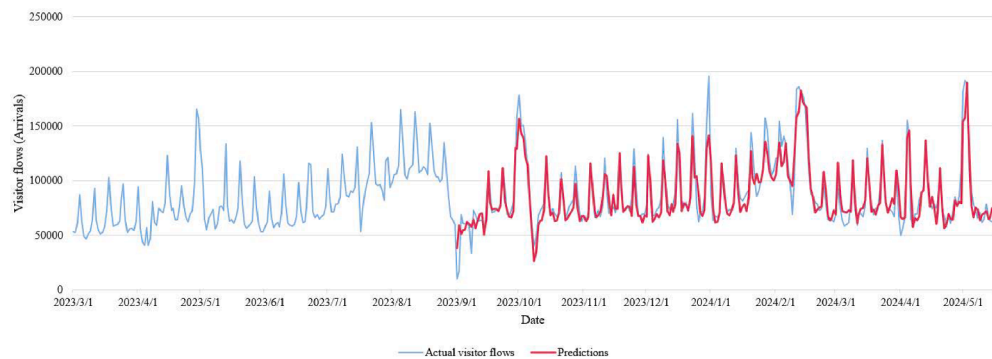


Fig. 8. GBDT rolling prediction on visitor flows under one-day horizon.

tourism seasons, which are associated with a high number of visitor flows. Adverse weather conditions, such as heavy rain, negatively impact cross-border visitor flows. The maximum temperature also exhibits high importance in the prediction, suggesting the seasonal fluctuation of visitor flows.

Visitors are also largely motivated by an event's popularity on social media platforms to embark on cross-city trips. The promotional popularity of music concerts, exhibitions, and fireworks displays positively impacts cross-border visitor flows. Among these events, increased promotional popularity for exhibitions has a smaller effect on attracting visitors compared to fireworks displays and music concerts. This variation arises because exhibitions are often held over a long period. Although we excluded events with more than 30 sessions during data processing, the remaining exhibitions still average six sessions, significantly more than the average of three sessions for other event types. The multi-session nature of exhibitions causes attendees to be distributed across many sessions, diluting the impact on daily visitor flows.

The findings shown in Fig. 9 provide valuable insights into the effect of word-of-mouth on different types of events. For music concerts, an increase in word-of-mouth popularity positively impacts visitor flows. However, for exhibitions, increases in word-of-mouth popularity do not exhibit a clear positive or negative impact on visitor numbers. Similar to the pattern observed for movie audiences, whose attendance typically decreases as the screening period progresses (Delre and Luffarelli, 2023), exhibitions generally run for extended durations, with visitor numbers often peaking during the opening days and subsequently declining to a low yet stable level. In addition, as observed in Table 4, social media engagement following the first session of an exhibition is also significantly lower compared to other event types. Furthermore, word-of-mouth popularity influences recurring events with multiple sessions but is not applicable to the first session. Specifically, for non-recurring events or the first session of recurring events, the word-of-mouth popularity feature is set to zero, which can limit feature variation and lead to depressed feature importance. This also highlights limitations of the word-of-mouth popularity metric, and future studies can explore alternative measurements.

The promotional popularity features of sports competitions are less important for predicting visitor flows. The word-of-mouth popularity feature has no observable impact and, therefore, is not represented in SHAP results. This indicates that the social media exposure of sports competitions in Hong Kong may not significantly attract visitors from Chinese Mainland. On one hand, this may be because these events are not as competitive as the high-profile international sports competitions held frequently in Chinese Mainland. On the other hand, some major sporting events offer live broadcasts, allowing individuals to participate by watching online or following the results on social media, rather than attending in person.

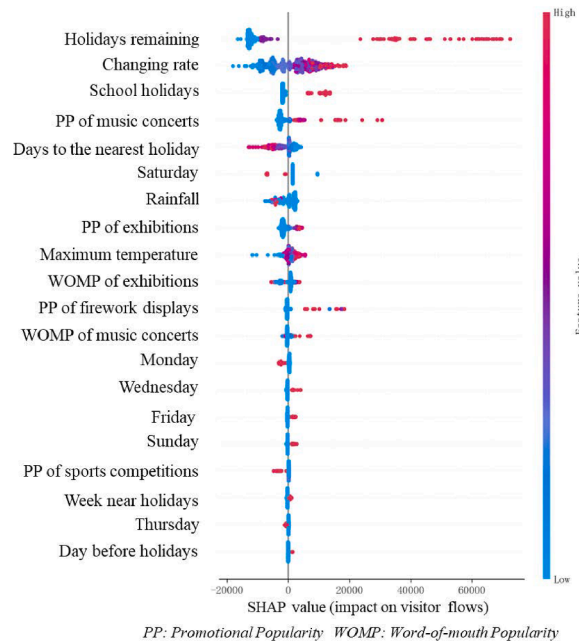


Fig. 9. Feature importance for visitors flow prediction.

5.5. Events' impact across travel modes

To demonstrate the practical relevance of our model for the efficient operation of cross-city infrastructure, we apply it to analyze Chinese Mainland visitor flows at three of Hong Kong's busiest border entry points. These entry points, each offering distinct transport services, are the Lok Ma Chau Spur Line (for metro service), West Kowloon HSR Station, and Hong Kong International Airport. Our findings indicate that the impact of events on visitor flows varies significantly across different entry points, and the predictive performance of our models also differs accordingly. The prediction accuracy for visitors entering through the Lok Ma Chau Spur Line and the West Kowloon HSR Station achieves R-squared values of 78.3% and 82.9%, respectively. In contrast, the predictive performance for air travelers is significantly lower, with an R-squared value of only 51.2%.

Fig. 10 presents the top eight features influencing visitor flows for each of the three entry points. As each mode of transportation caters to different travel distances, visitors passing through different entry points likely exhibit distinct profiles in terms of trip origin, travel purpose, and level of itinerary flexibility. Notably, events' social media popularity features do not exhibit a significant impact on visitor arrivals through air travel, which are more strongly associated with holidays. This may be attributed to the fact that airport arrivals generally originate from more distant cities. Such trips are usually planned well in advance and incur higher travel costs, and thus are typically intended for vacation or business purposes over multiple days or weeks. Most events, however, only last for a few hours and are often not the primary attractors for such trips.

Visitor flows through both the metro and HSR entry points are positively associated with the promotional popularity of events such as concerts and fireworks displays. In particular, visitors arriving by metro are more responsive to the word-of-mouth effects of concerts. Note that the metro entry points allow residents in Shenzhen, a city bordering Hong Kong to the north with a population of 18 million, to travel to Hong Kong through public transportation with low costs and high flexibility, thereby encouraging them to undertake short-term visits for events in Hong Kong. Consequently, they are more likely to attend subsequent concerts after observing increased social media popularity.

In contrast, travelers arriving at the HSR entry point are primarily from cities within the GBA and nearby regions. The shortened travel time provided by the HSR benefits visitors traveling longer distances, such as those from Guangzhou, Dongguan, Hunan, and Wuhan. Compared to metro travelers, HSR passengers are required to book their tickets and plan their trips in advance. As a result, they are less influenced by word-of-mouth about events when planning cross-city trips. The heterogeneous impacts of events on visitor flows via different transport modes provide valuable insights for event-aware intercity transport infrastructure planning and management strategies.

6. Policy implications: a case study of Hong Kong coliseum

To illustrate how event-aware analysis of visitor flows can inform targeted policy interventions regarding border operations and transportation management during major events, we present a case study focusing on a major concert venue in Hong Kong, namely *Hong Kong Coliseum* (also known as the Hung Hom Coliseum), which is centrally located in the city. For concert events, transportation

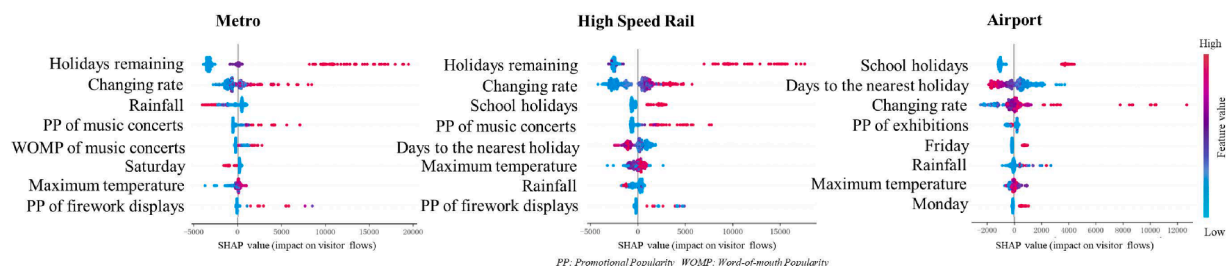


Fig. 10. Feature importance on visitors flow by border entry points (Top 8 features).



Fig. 11. Representative social media posts regarding travel experiences after concerts at the Hong Kong Coliseum.

authorities can often utilize ticket sales data to estimate the expected audience size and implement appropriate traffic control measures around the venue. However, the framework developed in this study can further enable them to estimate the number of cross-city visitors among the audience, which can support event-aware policymaking in several ways.

First, by examining the association between the concert popularity on social media and the volume of visitors passing through different border checkpoints, local authorities can effectively identify which checkpoints are susceptible to disruptions caused by surges in event-driven visitor flows. Our results demonstrate that the social media popularity of concerts at Hong Kong Coliseum significantly influences the number of cross-border visitor flows through certain checkpoints, including Lok Ma Chau Spur Line, Lo Wu, West Kowloon HSR Station, Shenzhen Bay, and China Ferry Terminal. Both the promotional popularity and word-of-mouth popularity of concerts are among the top predictors of cross-border visitor flows through these checkpoints. When these popularity

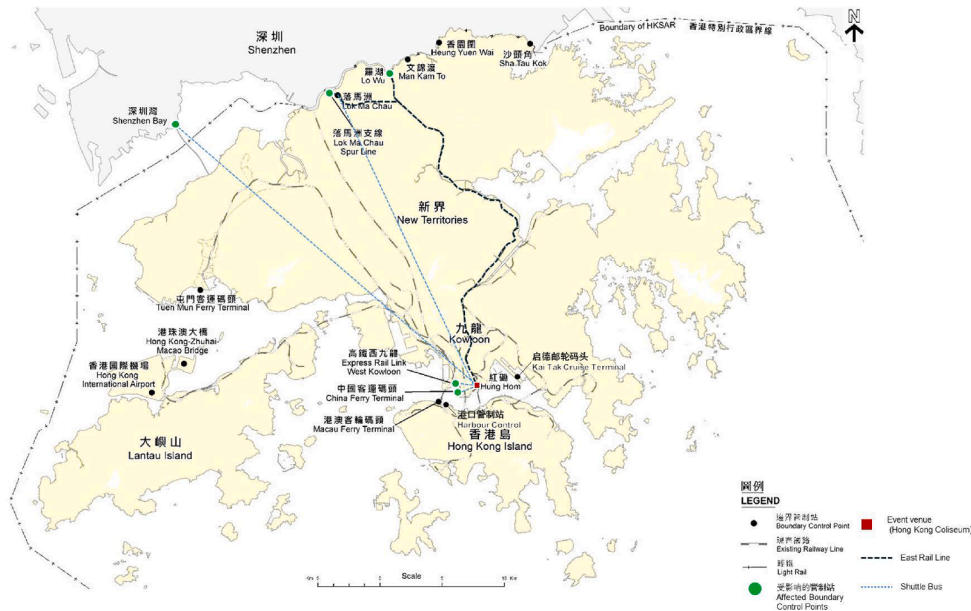


Fig. 12. Illustration of special transportation service arrangements for Chinese Mainland visitors attending concerts at Hong Kong Coliseum.

features are integrated with other conventional variables in *Feature Set 1*, the performance of visitor flow prediction models can be further enhanced, with the R^2 values exceeding 73% for Lok Ma Chau Spur Line, 76% for Shenzhen Bay, 79% for Lo Wu, 80% for West Kowloon HSR Station, and 81% for China Ferry Terminal. Based on the anticipated visitor flows, targeted measures can be implemented to mitigate the potential disruptions at these checkpoints. For example, many concerts end after the operating hours of some of the most accessible border checkpoints, limiting visitors' options to return to their origin cities. Therefore, based on the event's venue, schedules, and its potential appeal to visitors, local authorities may extend the operating hours of relevant checkpoints and deploy additional customs personnel, which can help accommodate the anticipated surges in visitor flows and ensure efficient cross-border mobility during event periods.

Second, public transportation service adjustments and special shuttle services are often needed to accommodate the event-driven surges in travel demand between the event venue and key border checkpoints as well as major transportation hubs, thereby mitigating the potentially disruptive impacts on local transportation systems. Since local transportation systems are typically designed to serve regular travel demand (e.g., commuting), they may be vulnerable to sudden surges caused by popular events, especially when large numbers of visitors try to return to their origin cities after the concerts end late at night. As illustrated in Fig. 11, cross-city concertgoers frequently complain on social media about the inadequate bus service capacity and low frequency at Hong Kong Coliseum at late night, which often result in long waiting times and considerable inconvenience for visitors. To address these challenges, transportation authorities can adjust relevant public transportation services, targeting those connecting the event venue to popular border checkpoints. For example, Fig. 12 illustrates a number of key routes between Hong Kong Coliseum (indicated by the red square) and primary border checkpoints likely used by cross-city eventgoers (marked with green circles). Hung Hom station, served by the East Rail Line (shown as the dashed black line), provides direct connections between the Hong Kong Coliseum and both the Lo Wu and Lok Ma Chau Spur Line checkpoints. Therefore, it can be necessary to increase the service frequency or extend operating hours to adequately accommodate the demand surge during event periods. In addition, special shuttle services can be established to serve travel demand between the event venue and key border checkpoints and major transportation hubs (represented in dashed blue lines in Fig. 12). Specifically for concerts in the Hong Kong Coliseum, local authorities may provide long-distance shuttle services directly to major land checkpoints such as Shenzhen Bay and Lok Ma Chau, as well as short-distance services to nearby transportation hubs such as West Kowloon HSR Station or China Ferry Terminal. The specific service frequency and capacity can be designed based on event schedules and anticipated visitor flows, thereby ensuring efficient arrivals of visitors before events start and departures right after they end.

Third, effective on-site traffic and crowd management strategies are also essential for large-scale events to ensure public safety and maintain efficient mobility around the event venue. Unlike local eventgoers, cross-city visitors are generally less familiar with the local environment and would require more guidance in accessing and egressing the event area. With the proposed visitor flow prediction method, transportation authorities can gain sufficient lead time to collaborate closely with event organizers, parking facility managers, transport police and emergency service agencies to implement coordinated management plans. These measures may include designing dedicated pick-up/drop-off areas for special shuttle services near the event venue, establishing temporary pedestrian zones, providing crowd navigation information, implementing traffic re-routing strategies, optimizing parking management, and enhancing

surveillance systems. Collectively, these efforts are designed to ensure the safety of event participants and the general public, while maintaining efficient transportation operations around the event venue.

7. Conclusion and discussion

7.1. Key findings and contributions

This study addresses the gap in understanding how diverse and concurrent events impact cross-city visitor flows. By leveraging the robust text-processing and semantic analysis capabilities of LLMs, we compile city-level event datasets and enable the scalable utilization of user-generated social media content for event-related dynamics. Social media's promotional and word-of-mouth effects significantly influence people's intentions to undertake cross-city trips to attend events (Hu et al., 2022; Cui et al., 2018; Duan et al., 2008; Dellarocas et al., 2007). To our knowledge, this study is the first to quantify event popularity by integrating both pre-event promotion and post-event word-of-mouth effects, subsequently applying these features to predict cross-city visitor flows. By incorporating these popularity features, the study proposes an event-aware machine learning model that effectively enhances the accuracy of visitor flow predictions.

Through a case study of Hong Kong, we further reveal heterogeneous effects across event types and identify key travel modes that are susceptible to disruptions caused by events. An application scenario focused on a major concert venue demonstrates the practical implications of event-aware transport service provision and targeted policy formulation. Importantly, the generalizability of this approach allows its application in other regions experiencing similar event-motivated cross-city travel dynamics.

7.2. Broader practical implications

This study offers significant practical implications for managing transport systems under diverse and concurrent events. Responsive management requires coordinated policy measures to ensure efficient transport operations at multiple levels. Timely and localized information on predicted visitor flows and the potential impacts of events should be shared among all relevant stakeholders, including border management authorities, public transportation operators, parking facility managers, transport police and emergency service agencies, and event organizers. By sharing this information, stakeholders can collaboratively develop strategies to address the challenges posed by event-driven travel demand. From a long-term planning perspective, the integration of the event-aware method into regional travel demand forecasting can provide valuable insights for the development and enhancement of cross-city transport infrastructure, which is typically designed to accommodate regular demand patterns.

From a broader perspective, these analytical results can help foster cross-departmental coordination and amplify the collective economic benefits. Understanding the impact of various events and improving the prediction of visitor flows are essential for broader event planning, tourism development, and destination branding strategies. In the case of Hong Kong, SHAP analysis demonstrates that events such as music concerts, fireworks displays, and exhibitions with high promotional popularity are closely associated with increased visitor demand. Destination branding and tourism authorities can optimize event planning strategies to prioritize events with greater visitor appeal or coordinate event portfolios (Ziakas and Getz, 2021; Ziakas, 2020). Event portfolio management allows concurrent events to generate a synergistic effect, providing a richer and more diverse array of attractions that appeal to wider visitor groups. Additionally, the hospitality and retail sectors can leverage these predictive insights to make informed decisions regarding inventory management, staffing, and dynamic pricing strategies. By anticipating surges in visitor demand during high-profile events, businesses can optimize operations, enhance customer experience, and maximize profitability (Wu et al., 2024). Considering the potential origins of visitors using different travel modes, the study highlights the varied geographical impact of pre-event promotions and post-event word-of-mouth communication. These insights help event and destination marketing organizations optimize the allocation of online marketing resources, enabling them to effectively target key audiences, enhance advertising efficiency, and maximize revenue.

7.3. Limitations and future research

Although this study provides a novel framework for event-aware analysis, several limitations warrant acknowledgment.

The first set of limitations concerns the nature of social media data. Although we define event popularity through social media engagement, it is important to note that users who “like” or “collect” posts do not necessarily plan to attend the events (Önder et al., 2020). Even among those who express their intention to do so, there is often a significant gap between intention and actual attendance behavior (Lee et al., 2014). Therefore, in our study, social media popularity primarily reflects an event's exposure rather than serving as a reliable measure of actual attendance. Furthermore, our analysis of social media popularity mainly utilizes the platform *Rednote*, a popular Chinese social media network. Although the platform's main user demographic aligns with key groups of event visitors and cross-border visitors, *Rednote* has a higher proportion of young and female users. This may slightly skew the representation of the true demographic distribution of event visitors.

To address these biases, future research should implement a multi-platform comparative approach to harmonize demographic differences. Moreover, incorporating individual mobility data, such as smart card transactions or mobile phone traces, would allow researchers to better understand visitors' spatiotemporal mobility patterns during events. This data would enable a more detailed analysis of event-motivated human mobility within a city. In addition, individual mobility data can be used to estimate the temporal flow of visitors around specific event venues. This allows for a more nuanced understanding of how social media popularity is related

to spatiotemporal travel demand patterns throughout the event's entire lifecycle. In addition, future research could investigate the economic impact of cross-city mobility motivated by diverse events. Building on the findings regarding the effectiveness of different events and their online popularity, researchers can further evaluate the overall economic benefits of hosting different types of events.

A second category of limitations involves the deployment of LLMs. The reliability of the observed results is intrinsically tied to model-version dependency; while GPT-4 represented the state-of-the-art during the data-processing phase, it remains susceptible to “hallucinations”, the generation of factually incorrect or inconsistent information (Bang et al., 2023; Liang et al., 2024). In this study, semantic analysis relies on human evaluation as a benchmark, which is inherently characterized by cognitive subjectivity. This subjectivity complicates the definitive measurement of divergence between LLM outputs and an absolute “ground truth”, a challenge that distinguishes this approach from strictly deterministic quantitative models. To address these evaluative challenges, recent advancements have introduced the “LLM-as-a-Judge” framework (Zheng et al., 2023) as a scalable alternative to human evaluation. Additionally, the implementation of agentic reflection patterns (Shinn et al., 2023) where models iteratively critique and refine their own outputs, represents a promising future direction to improve reliability.

Despite the prompt sensitivity analyses detailed in Appendix B, LLM outputs may exhibit fluctuations contingent upon specific phrasing and the granularity of instructions. Furthermore, the iterative nature of API-based prompting remains resource-intensive, incurring significant computational latency and monetary costs when applied to large-scale datasets. However, the rapid evolution of the field, particularly the emergence of the Model Context Protocol (MCP) and skill-based agentic workflows (Hou et al., 2025; Jiang et al., 2026), suggests a transition to increasingly stable, automated, and interconnected data mining pipelines. By adopting these burgeoning agentic frameworks and newer model iterations, future research may further enhance computational efficiency and lower the technical barriers for local authorities seeking to implement such event-aware analytical frameworks.

CRedit authorship contribution statement

Xiaohan Wang: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization; **Zhan Zhao:** Writing – original draft, Methodology, Formal analysis, Conceptualization; **Ruiyu Wang:** Writing – review & editing, Data curation; **Yang Xu:** Writing – review & editing, Methodology.

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Appendix A. LLM performance in semantic analysis

To assess the reliability of the LLM-based approach for semantic analysis, we compare the results of event-related social media posts identified by LLMs with those obtained through manual screening, as used in previous research. Specifically, we implement a stratified sampling strategy by randomly selecting 5% of events from each event category. A total of 459 social media posts are sampled. The relevance of social media posts to these sampled events is then manually evaluated. Three researchers with backgrounds in transportation and urban planning are involved in manual labeling. Each post is independently evaluated by two researchers to determine its relevance to the specific event. In cases of disagreement, a third researcher provides a final adjudication. The process takes approximately 11 person-hours to complete.

The results indicate that the LLM achieves 93% agreement with human judgments. As presented in [Table A.1](#), discrepancies occur primarily in cases where social media posts contain limited textual descriptions and the model has to mainly rely on visual information, such as images and videos (see Case 1). Further inconsistencies emerge in cases where an absolute objective truth is absent. Such scenarios, including posts with multiple event recommendations (see Case 2), are equally challenging for human evaluators.

A major challenge for conventional NLP models to identify relevant social media posts from noisy raw data is the disparity in linguistic style and register between event information and user-generated content. To further demonstrate the superior performance of LLMs over conventional NLP methods, we compare the LLM with Sentence-BERT, which is an adaptation of the BERT architecture designed for efficient sentence similarity computation and clustering by encoding entire sentences into dense vector representations. For Sentence-BERT, a similarity threshold of 50% is used to determine relevance. The comparative results are summarized in [Table A.2](#), where accuracy, precision, recall, and F1 score metrics are used to assess the alignment of model results with human judgments. The LLM demonstrates significantly better performance than Sentence-BERT across all metrics. In identifying event-related social media posts, the LLM-based method achieves greater accuracy and stronger alignment with human evaluators compared to Sentence-BERT.

Appendix B. LLM prompting sensitivity

Previous studies have indicated that minor wording changes in prompts can influence LLM performance (Shi et al., 2024; Voronov et al., 2024; Polo et al., 2024; Sclar et al., 2024), including spurious template features, slight lexical variations, and paraphrasing. In this study, to assess the sensitivity of *Prompt 4* to variations in prompt wording, we use LLMs to generate three alternative versions of the prompt for identifying the relevance of social media posts. Variation 1 is more concise compared to the original prompt. Variation 2 maintains a similar list structure to the original but simplifies the question with less context. Variation 3 employs a task decomposition approach, providing more detailed instructions to the model. The performance results of these three variations are

Table A.1
Inconsistent cases in semantic analysis.

Cases	Event information	Social media post	LLM judgment	Human judgment
1	Title: Labor Day Fireworks Type: Fireworks Description: The Labor Day holiday fireworks show will last for 10 minutes. The fireworks will be launched from a barge 130 meters offshore, reaching up to about 100 meters above the water. The performance will feature a golden theme, with giant “HK” letters and smiling face patterns blooming over the Victoria Harbour skyline. Combined with the A Symphony of Lights multimedia light show, it will offer a dual visual experience to attract spectators.	Title: Record of the Hong Kong Light Show Content: These are the fireworks I saw — I arrived late and couldn’t get a good spot. Geo-tags: None Hashtags: #LaborDayInHongKong	False	True
2	Title: Rex Orange County Asia Tour Concert – Hong Kong 2023 Type: Concerts Description: Rex Orange County will hold a series of concerts across multiple cities in Asia, with the Hong Kong show taking place on October 10. The performance will feature popular songs from his new album Who Cares?, including OPEN A WINDOW, a collaboration with Tyler, The Creator. This tour marks Rex’s first solo concerts in Asia, continuing the success of his North American tour.	Title: Events in Hong Kong this September that you must not miss — you’ll regret it if you do. Content: How vibrant are Hong Kong’s events? If you are visiting in September, you definitely should not miss these: 1. Hungry Ghost Festival Cultural Fair In addition to exhibitions, games, performances, and workshops related to the Yu Lan Festival, this year’s event also features role-playing mystery games and virtual reality tours of the Ghost Festival, allowing visitors to experience and understand the festival’s cultural significance through technology. Date: September 1–3, 2023 Location: Court No. 1, Victoria Park 2. 2023 Hong Kong Open Badminton Championships This event is part of the BWF World Tour Super 500 series. It gathers top badminton players from more than 20 countries and regions to compete for glory and qualification spots for the 2024 Paris Olympics. Date: September 12–17, 2023 Location: Hong Kong Coliseum 3. Fire Dragon Dance at Pok Fu Lam and Tai Hang During the Mid-Autumn Festival, Pok Fu Lam Village on Hong Kong Island holds its annual Fire Dragon Dance and the traditional “Dragon Returns to the Sea” ritual. In Tai Hang, Causeway Bay, residents perform the traditional Fire Dragon Dance, beating drums and gongs as they lead a 67-meter-long dragon covered in incense sticks through the streets, transforming the usually quiet neighborhood into a lively celebration. Date: Pok Fu Lam Village – September 28–29, 2023; Tai Hang – September 28–30, 2023 Location: Pok Fu Lam Village (best viewing spot: village entrance); Tai Hang, Causeway Bay (best viewing spot: Wun Sha Street) 4. Radiance of Gold: Ancient Gold from the Hong Kong Palace Museum and Mengdiexuan Collection This exhibition features gold ornaments from the Eurasian Steppe, Tubo (Tibet), and Central Plains regions, dating back as early as the 18th century BCE. Exhibits include ornate jewelry worn by ancient nobles and decorative items for horses and carriages. Date: February 22 – October 3, 2023 Location: Gallery 9, Hong Kong Palace Museum, West Kowloon Cultural District 5. Hong Kong Trade Development Council (HKTDC) Events Hong Kong Watch & Clock Fair 2023 (Physical Exhibition) Salon de TE 2023 (Physical Exhibition) CENTRESTAGE 2023 – Hong Kong International Fashion Fair 6. Concerts A total of eleven concerts will be held, featuring artists such as Waa Wei, Post Malone, and KOLOR. In October and November, Charlie Puth, Sam Smith, Rex Orange County, and A-Mei will also hold concerts in Hong Kong. Music fans should not miss these performances. Geo-tags: None Hashtags: #Hong Kong life, #Hong Kong concerts, #Hong Kong events, #Hungry Ghost Festival (Yu Lan Festival), #concerts, #postmalone, #CharliePuth, #samsmith	True	False

Table A.2
Performance evaluation of LLM and Sentence-BERT.

Metrics	LLM (GPT-4)	Sentence-BERT
Accuracy (%)	93.25	62.31
Precision (%)	99.16	65.75
Recall (%)	89.06	72.45
F1 score (%)	93.84	68.94

Note that the metrics are computed based on the agreements with human judgments, which do not constitute the true ground truth.

presented in Table B.1. Variations 1 and 2 exhibit lower performance compared to the original prompt, whereas Variation 3, using task decomposition, achieves results comparable to the original prompt. These findings suggest that LLMs can achieve satisfactory performance under various prompt formatting conditions, attaining over 88% accuracy. Providing a structured and clear context along with instructions can further enhance model performance. Nevertheless, excessively elaborate prompts or instructions do not necessarily lead to additional improvements.

Table B.1
Prompt sensitivity analysis of LLM performance.

Metrics	Prompt Variation 1	Prompt Variation 2	Prompt Variation 3
Prompt Text	Compare the event and social media post below. Answer “Yes” if the post is related to the event, or “No” if not. Event: {event title} ({event type}), {event description} Post: {post title}, {post content} Location: {post geo-tags} Tags: {post tags} The post must be about an event in Hong Kong with a clear connection to the described event.	Task: Determine if a social media post relates to the given event. Event Details: Title: {event title} Type: {event type} Description: {event description} Post Details: Title: {post title} Content: {post content} Location: {post geo-tags} Tags: {post tags} Criteria: The post must mention an event in Hong Kong. The post must clearly connect to the event description. ANSWER (Yes/No):	Task: You are to analyze whether a social media post is related to a specific event. Step 1: Review the event information: Event Title: {event title} Event Type: {event type} Event Description: {event description} Step 2: Review the social media post: Post Title: {post title} Post Content: {post content} Location Tags: {post geo-tags} Hashtags: {post tags} Step 3: Apply these criteria: (1) Does the post mention an event in Hong Kong? (2) Is there semantic overlap between the post and event description? (3) Do temporal/spatial/thematic elements align? Step 4: Make your determination. Answer with exactly one word: “Yes” (if related) or “No” (if not related).
Accuracy (%)	88.02	88.89	93.03
Precision (%)	99.07	99.08	99.16
Recall (%)	80.00	81.51	88.68
F1 score (%)	88.52	89.44	93.63

Appendix C. Decay effect of word-of-mouth popularity

We assume that the online word-of-mouth generated by a specific event session potentially impacts all subsequent sessions, with the effect either remaining constant or decaying over time. Word-of-mouth popularity for session n is defined in Eq. (1), where n' denotes the index of a prior session within the same event. To model the influence of word-of-mouth generated by session n' , $WOM_{n'}$, we examine three influence patterns represented by the weight factor $\alpha_{n,n'}$. When the word-of-mouth effect is assumed to be evenly distributed across all subsequent sessions without decay, we set the weight factor to $\alpha_{n,n'} = 1$. In cases of exponential decay, the weight factor follows the definition in Eq. (2), where we test decay rates $\beta \in \{0.2, 0.4, 0.6, 0.8\}$. For the linear decay pattern, the weight factor is given by $\alpha_{n,n'} = \max(1 - \beta'(n - n'), 0)$, where β' represents the linear decay rate tested at values $\beta' \in \{0.05, 0.1, 0.2\}$.

By evaluating these different patterns and their corresponding parameters, we find that the model achieves optimal performance when modeling the online word-of-mouth effect as an exponential decay function with $\beta = 0.2$. This temporal decay pattern serves as the most representative measure of an event's word-of-mouth popularity.

Appendix D. Model details

D.1. GBDT model details

GBDT builds an additive model in a forward stage-wise manner. It optimizes an objective function that consists of a differentiable loss function and a regularization term. Let $\mathcal{L}(y_t, F(x_t))$ denote the loss function where y_t is the true value and $F(x_t)$ is the prediction. Here, x_t represents the feature vector for day t . The objective function J is defined as:

$$J = \min_J \sum_{t=1}^T w_t \mathcal{L}(y_t, F_{t-1}(x_t) + \gamma h_t(x_t)) + \Omega(h_t) \quad (\text{D.1})$$

where w_t denotes the sample weight for day t , T the total length of training data (number of days). $F_{t-1}(x_t)$ represents the prediction from the ensemble of trees up to the $(t - 1)^{\text{th}}$ iteration, $h_t(x_t)$ represents the prediction from the t^{th} new tree. γ is the learning rate, $\Omega(h_t)$ is the regularization term to penalize the complexity of the model.

D.2. Sample weighting mechanism

Given the rebound of tourism in the post-COVID era, recent trends provide more reliable indicators for forecasting future visitor flows. We introduce a sample weighting mechanism into the model, emphasizing recent dynamics. Three different sample weighting functions are tested, including linear decay, exponential decay, and power-law decay, each with various decay parameters δ .

For the linear decay function, the sample weight w_t for a given day t is defined in Eq. (5). We evaluate the model for δ values ranging from 0 to 0.006 at increments of 0.001.

For the exponential decay function, the sample weight w_t is given by:

$$w_t = e^{-\delta'(T-t-1)} \quad (\text{D.2})$$

where T denotes the final day in the training period. We examine δ' values in the range 0.005 to 0.025.

For the power-law decay function, the sample weight w_t is determined as follows:

$$w_t = \frac{1}{(1 + T - t)^{\hat{\delta}}} \quad (\text{D.3})$$

where the value of $\hat{\delta}$ is examined over the range from 0.025 to 0.15.

By systematically comparing the performance of these three sample weighting functions across all tested decay factors, we find that the linear decay function with the decay parameter ($\delta = 0.005$) yields the optimal predictive performance. This approach enables the model to adaptively emphasize recent data, effectively capturing the evolving trends in post-pandemic tourism flows.

D.3. Hyperparameters of models

We perform a comprehensive grid search to determine the optimal hyperparameters for GBDT model. As detailed in Table D.1, we systematically adjust the learning rate, maximum depth, the number of estimators (trees), min samples split, and min samples leaf. This grid search identifies those hyperparameters that result in the best performance as measured by the R^2 value. In the GBDT model, the hyperparameters are optimized as follows: a learning rate of 0.08, a maximum tree depth of 2300 estimators, a minimum samples split of 5, and a minimum samples per leaf of 3.

We compare the GBDT model against six commonly used baseline models: XGBoost, CatBoost, LightGBM, RF, LR, and ARIMA. To ensure optimal performance for each baseline model, we apply a grid search on the LightGBM, CatBoost, XGBoost, and RF models to determine the best hyperparameters, including learning rate, maximum depth, number of estimators, etc. We also implement the same rolling prediction and sample weighting mechanisms in these models. For the ARIMA model, we use a grid search to optimize its specific parameters: autoregression order p , degree of differencing d , and moving average order q . The search steps and the best hyperparameters for the models are listed in the Table D.1.

Table D.1
Grid search for best hyperparameters.

Model	Hyperparameters	Steps	Best hyperparameters
GBDT	Learning rate	0.01, 0.05, 0.08, 0.1	0.08
	Maximum depth	2, 3, 5, 7	2
	Number of estimators	100, 300, 500, 1000	300
	Min samples split	2, 5	5
	Min samples leaf	1, 3	3
XGBoost	Learning rate	0.01, 0.05, 0.08, 0.1	0.08
	Maximum depth	2, 3, 5, 7	2
	Number of estimators	100, 300, 500, 1000	500
	Minimum weights in a leaf	3, 5	3
CatBoost	Learning rate	0.01, 0.05, 0.08, 0.1	0.1
	Maximum depth	2, 3, 5, 7	3
	Iterations	100, 300, 500, 1000	300
	L2 regularization	1, 3	1
	min data in leaf	3, 10	10
LightGBM	Learning rate	0.01, 0.05, 0.08, 0.1	0.08
	Maximum depth	2, 3, 5, 7	3
	Number of estimators	100, 300, 500, 1000	300
	Number of leaves	15, 31	31
	Min child samples	3, 5	5
RF	Maximum depth	2, 3, 5, 7	7
	Number of estimators	100, 300, 500, 1000	100
	Min samples split	2, 5	5
	Min samples leaf	1, 3	1
Arima	p	1, 7, 14, 28	1
	d	1	1
	q	1, 7	7

References

- Bang, Y., Cahyawijaya, S., Lee, N., Dai, W., Su, D., Wilie, B., Lovenia, H., Ji, Z., Yu, T., Chung, W., Do, Q.V., Xu, Y., Fung, P., 2023. A multitask, multilingual, multimodal evaluation of ChatGPT on reasoning, hallucination, and interactivity. <https://doi.org/10.48550/arXiv.2302.04023>
- Becken, S., 2013. Measuring the effect of weather on tourism: a destination- and activity-based analysis. *J. Travel Res.* 52 (2), 156–167. <https://journals-sagepub-com.eproxy.lib.hku.hk/doi/full/10.1177/0047287512461569>.
- Boivin, M., Tanguay, G.A., 2019. Analysis of the determinants of urban tourism attractiveness: the case of Québec City and Bordeaux. *J. Destination Mark. Manage.* 11, 67–79. <https://www.sciencedirect.com/science/article/pii/S2212571X16303560>.
- Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., Amodei, D., 2020. Language models are few-shot learners. [arXiv:2005.14165](https://arxiv.org/abs/2005.14165) [cs].
- Chang, J.-J., Chen, R.-F., Lin, C.-L., et al., 2022. Exploring the driving factors of urban music festival tourism and service development strategies using the modified SIA-NRM approach. *Sustainability* 14 (12), 7498. <https://www.mdpi.com/2071-1050/14/12/7498>.
- Chen, E., Ye, Z., Wang, C., Xu, M., et al., 2020. Subway passenger flow prediction for special events using smart card data. *IEEE Trans. Intell. Transp. Syst.* 21 (3), 1109–1120. <https://ieeexplore.ieee.org/abstract/document/8725576>.
- Cui, R., Gallino, S., Moreno, A., Zhang, D.J., 2018. The operational value of social media information. *Prod. Oper. Manage.* 27 (10), 1749–1769. <https://doi.org/10.1111/poms.12707>.
- De Montis, A., Barthélemy, M., Chessa, A., Vespignani, A., et al., 2007. The structure of interurban traffic: a weighted network analysis. *Environ. Plann. B Plann. Des.* 34 (5), 905–924. <https://doi.org/10.1068/b32128>.
- De Montis, A., Chessa, A., Campagna, M., Caschili, S., Deplano, G., et al., 2010. Modeling commuting systems through a complex network analysis: a study of the Italian islands of Sardinia and Sicily. *J. Transp. Land Use* 2 (3/4), 39–55. <https://www.jstor.org/stable/26201637>.
- Dedeoğlu, B.B., Taheri, B., Okumus, F., Gannon, M., 2020. Understanding the importance that consumers attach to social media sharing (ISMS): scale development and validation. *Tourism Manage.* 76, 103954. <https://doi.org/10.1016/j.tourman.2019.103954>.
- Dellarocas, C., Zhang, X.M., Awad, N.F., 2007. Exploring the value of online product reviews in forecasting sales: the case of motion pictures. *J. Interact. Mark.* 21 (4), 23–45. <https://doi.org/10.1002/dir.20087>.
- Delre, S.A., Luffarelli, J., 2023. Consumer reviews and product life cycle: on the temporal dynamics of electronic word of mouth on movie box office. *J. Bus. Res.* 156, 113329. <https://www.sciencedirect.com/science/article/pii/S0148296322007949>.
- Devlin, J., Chang, M.-W., Lee, K., Toutanova, K., 2019. BERT: pre-training of deep bidirectional transformers for language understanding. <https://doi.org/10.48550/arXiv.1810.04805>.
- Du, R., Yu, Z., Mei, T., Wang, Z., Wang, Z., Guo, B., et al., 2014. Predicting activity attendance in event-based social networks: content, context and social influence. In: *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. ACM, Seattle Washington, pp. 425–434. <https://dl.acm.org/doi/10.1145/2632048.2632063>.
- Duan, W., Gu, B., Whinston, A.B., 2008. Do online reviews matter? – an empirical investigation of panel data. *Decis. Support Syst.* 45 (4), 1007–1016. <https://www.sciencedirect.com/science/article/pii/S0167923608000754>.
- Eisenstein, J.H., 2013. What to do about bad language on the internet. In: *Vanderwende, L., Daumé, III, H., Kirchhoff, K. (Eds.), Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Atlanta, Georgia, pp. 359–369. <https://aclanthology.org/N13-1037/>.
- Friedman, J.H., 2001. Greedy function approximation: a gradient boosting machine. *Ann. Stat.* 29 (5), 1189–1232. <http://www.jstor.org/stable/2699986>.
- Fu, J., Han, H., Su, X., Fan, C., et al., 2024. Towards human-AI collaborative urban science research enabled by pre-trained large language models. *Urban Inf.* 3 (1), 8. <https://doi.org/10.1007/s44212-024-00042-y>.
- Getz, D., Page, S.J., 2016. Progress and prospects for event tourism research. *Tourism Manage.* 52, 593–631. <https://www.sciencedirect.com/science/article/pii/S0261517715000679>.
- Heller, L.R., Stephenson, E.F., 2021. How does the super bowl affect host city tourism? *J. Sports Econ.* 22 (2), 183–201. <https://doi.org/10.1177/1527002520959393>.
- Hong Kong Planning Department, 2017. Northbound southbound: cross-boundary travel survey 2017. https://www.pland.gov.hk/pland_en/p_study/comp/s/nbsb2017/NBSB2017.pdf.
- Hong Kong Planning Department, 2021. Northbound southbound: cross-boundary travel survey 2021. https://www.pland.gov.hk/pland_en/p_study/comp/s/nbsb2021/NBSB2021.pdf.
- Hou, X., Zhao, Y., Wang, S., Wang, H., 2025. Model context protocol (MCP): landscape, security threats, and future research directions. [arXiv:2503.23278](https://arxiv.org/abs/2503.23278) [cs].
- Hu, M., Li, H., Song, H., Li, X., Law, R., 2022. Tourism demand forecasting using tourist-generated online review data. *Tourism Manage.* 90, 104490. <https://www.sciencedirect.com/science/article/pii/S0261517722000036>.
- Hudson, S., Roth, M.S., Madden, T.J., Hudson, R., 2015. The effects of social media on emotions, brand relationship quality, and word of mouth: an empirical study of music festival attendees. *Tourism Manage.* 47, 68–76. <https://www.sciencedirect.com/science/article/pii/S026151771400171X>.
- Jiang, J.J., Conrath, D.W., 1997. Semantic similarity based on corpus statistics and lexical taxonomy. <https://doi.org/10.48550/arXiv.cmp-lg/9709008>.
- Jiang, Y., Li, D., Deng, H., Ma, B., Wang, X., Wang, Q., Yu, G., 2026. SoK: agentic skills – beyond tool use in LLM agents. [arXiv:2602.20867](https://arxiv.org/abs/2602.20867) [cs].
- Lee, C.-K., Mjelde, J.W., Kim, T.-K., Lee, H.-M., 2014. Estimating the intention-behavior gap associated with a mega event: the case of the Expo 2012 Yeosu Korea. *Tourism Manage.* 41, 168–177. <https://www.sciencedirect.com/science/article/pii/S026151771300174X>.
- Lei, W.S.C., Chen, Z., Zhou, X., King, B., 2023. Organic destination imagery and social media: mapping tourism dynamics across China's greater bay area cities. *J. China Tourism Res.* 19 (2), 197–220. <https://doi.org/10.1080/19388160.2022.2047857>.
- Li, H., Gao, H., Song, H., 2023. Tourism forecasting with granular sentiment analysis. *Ann. Tourism Res.* 103, 103667. <https://www.sciencedirect.com/science/article/pii/S0160738323001408>.
- Li, T., Wang, J., Huang, J., Gao, X., et al., 2020. Exploring temporal heterogeneity in an intercity travel network: a comparative study between weekdays and holidays in China. *J. Geogr. Sci.* 30 (12), 1943–1962. <https://link.springer.com/article/10.1007/s11442-020-1821-9>.
- Liang, Y., Liu, Y., Wang, X., Zhao, Z., 2024. Exploring large language models for human mobility prediction under public events. *Comput. Environ. Urban Syst.* 112, 102153. <https://www.sciencedirect.com/science/article/pii/S0198971524000826>.
- Liu, X., Liu, A., Chen, J.L., Li, G., 2023. Impact of decomposition on time series bagging forecasting performance. *Tourism Management* 97, 104725. <https://www.sciencedirect.com/science/article/pii/S0261517723000079>.
- Lu, Q.-C., Zhang, J., Peng, Z.-R., Rahman, A.S., 2014. Inter-city travel behaviour adaptation to extreme weather events. *J. Transp. Geogr.* 41, 148–153. <https://www.sciencedirect.com/science/article/pii/S0966692314001781>.
- Mai, G., Huang, W., Sun, J., Song, S., Mishra, D., Liu, N., Gao, S., Liu, T., Cong, G., Hu, Y., Cundy, C., Li, Z., Zhu, R., Lao, N., et al., 2024. On the opportunities and challenges of foundation models for GeoAI (vision paper). *ACM Trans. Spatial Algorithms Syst.* 10 (2), 11:1–11:46. <https://doi.org/10.1145/3653070>.
- Mihalcea, R., Corley, C., Strapparava, C., 2006. Corpus-based and knowledge-based measures of text semantic similarity. *Proceedings of the 21st National Conference on Artificial Intelligence (AAAI'06)*, AAAI Press, Boston, Massachusetts, 1, 775–780.
- Ni, M., He, Q., Gao, J., 2016. Forecasting the subway passenger flow under event occurrences with social media. *IEEE Trans. Intell. Transp. Syst.*, 1–10. <http://ieeexplore.ieee.org/document/7583675>. <https://doi.org/10.1109/TITS.2016.2611644>.
- OpenAI, 2023. GPT-4 technical report. <https://api.semanticscholar.org/CorpusID:257532815>.
- Önder, I., Gunter, U., Gindl, S., et al., 2020. Utilizing Facebook statistics in tourism demand modeling and destination marketing. *J. Travel Res.* 59 (2), 195–208. <https://doi.org/10.1177/0047287519835969>.
- Pereira, F., Rodrigues, F., Polisciuc, E., Ben-Akiva, M., et al., 2015. Why so many people? Explaining nonhabitual transport overcrowding with internet data. *IEEE Trans. Intell. Transp. Syst.* 16 (3), 1370–1379. <http://ieeexplore.ieee.org/document/7021960/>.

- Pike, S., Page, S.J., 2014. Destination marketing organizations and destination marketing: a narrative analysis of the literature. *Tourism Manage.* 41, 202–227. <https://www.sciencedirect.com/science/article/pii/S0261517713001714>.
- Polo, F.M., Xu, R., Weber, L., Silva, M., Bhardwaj, O., Choshen, L., Oliveira, A. F. M.d., Sun, Y., Yurochkin, M., 2024. Efficient multi-prompt evaluation of LLMs. *arXiv:2405.17202* [cs].
- Reimers, N., Gurevych, I., et al., 2019. Sentence-BERT: sentence embeddings using Siamese BERT-networks. In: Inui, K., Jiang, J., Ng, V., Wan, X. (Eds.), *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Association for Computational Linguistics, Hong Kong, China, pp. 3982–3992. <https://doi.org/10.18653/v1/D19-1410>
- Rodrigues, F., Borysov, S.S., Ribeiro, B., Pereira, F.C., et al., 2018. A Bayesian additive model for understanding public transport usage in special events. *arXiv:1812.08755* [cs,stat]. <https://doi.org/10.1109/TPAMI.2016.2635136>
- Sclar, M., Choi, Y., Tsvetkov, Y., Suhr, A., 2024. Quantifying language models' sensitivity to spurious features in prompt design or: how I learned to start worrying about prompt formatting. *arXiv:2310.11324* [cs].
- Shi, C., Yang, K., Yang, J., Shen, C., 2024. Best arm identification for prompt learning under a limited budget. *arXiv:2402.09723* [stat] version:2.
- Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., Yao, S., 2023. Reflexion: language agents with verbal reinforcement learning. *arXiv:2303.11366v4*.
- Song, H., Zhang, H., 2024. Tourism demand modelling and forecasting: a horizon 2050 paper. *Tourism Rev.* Publisher: Emerald Publishing Limited. <https://www.emerald.com/insight/content/doi/10.1108/tr-12-2023-0882/full/html>.
- Tsui, K. W.H., Fu, X., Chen, T., Lei, Z., Wu, H., et al., 2021. Analyzing Hong Kong's inbound tourism: the impact of the COVID-19 pandemic. *IATSS Res.* 45 (4), 440–450. <https://www.sciencedirect.com/science/article/pii/S0386111221000443>.
- Voronov, A., Wolf, L., Ryabinin, M., 2024. Mind your format: towards consistent evaluation of in-context learning improvements. *arXiv:2401.06766* [cs].
- Wang, D.-g., Niu, Y., Qian, J., 2018. Evolution and optimization of China's urban tourism spatial structure: a high speed rail perspective. *Tourism Manage.* 64, 218–232. <https://www.sciencedirect.com/science/article/pii/S0261517717301863>.
- Wang, X., Pei, T., Li, K., Cen, Y., Shi, M., Zhuo, X., Mao, T., et al., 2022. Analysis of changes in population's cross-city travel patterns in the pre- and post-pandemic era: a case study of China. *Cities* 122, 103472. <https://www.sciencedirect.com/science/article/pii/S0264275121003711>.
- Wei, S., Pan, J., 2024. Spatiotemporal characteristics of residents' intercity travel in China under the impact of COVID-19 pandemic. *Cities* 152, 105206. <https://www.sciencedirect.com/science/article/pii/S0264275124004207>.
- Wirasinghe, S.C., Kumara, A.S., 1998. An aggregate demand model for intercity passenger travel in Sri Lanka. *Transportation* 25 (1), 77–98. <https://doi.org/10.1023/A:1004985506022>
- Wu, D.C., Zhong, S., Song, H., Wu, J., 2024. Do topic and sentiment matter? Predictive power of online reviews for hotel demand forecasting. *Int. J. Hospitality Manage.* 120, 103750. <https://www.sciencedirect.com/science/article/pii/S0278431924000628>.
- Xiang, Z., Gretzel, U., 2010. Role of social media in online travel information search. *Tourism Manage.* 31 (2), 179–188. <https://www.sciencedirect.com/science/article/pii/S0261517709000387>.
- Xue, G., Liu, S., Ren, L., Ma, Y., Gong, D., 2022. Forecasting the subway passenger flow under event occurrences with multivariate disturbances. *Expert Syst. Appl.* 188, 116057. <https://www.sciencedirect.com/science/article/pii/S0957417421013981>.
- Yamaguchi, H., Nakayama, S., 2023. Pattern analysis of Japanese long-distance travel change under the COVID-19 pandemic. *Transp. Res. Part A Policy Pract.* 176, 103805. <https://www.sciencedirect.com/science/article/pii/S0965856423002252>.
- Yang, X., Yue, X., Sun, H., Gao, Z., Wang, W., 2021. Impact of weather on freeway origin-destination volume in China. *Transp. Res. Part A Policy Pract.* 143, 30–47. <https://www.sciencedirect.com/science/article/pii/S0965856420307722>.
- Zeng, B., Gerritsen, R., 2014. What do we know about social media in tourism? A review. *Tourism Manage. Perspect.* 10, 27–36. <https://www.sciencedirect.com/science/article/pii/S2211973614000026>.
- Zhang, J., Jiang, W., Zhang, J., Wu, J., Wang, G., 2021. Exploring weather data to predict activity attendance in event-based social network: from the organizer's view. *ACM Trans. Web* 15 (2), 1–25. <https://dl.acm.org/doi/10.1145/3440134>.
- Zhang, T., Zhang, Z., Xue, G., et al., 2024. Mitigating the disturbances of events on tourism demand forecasting. *Ann. Oper. Res.* 342 (1), 1019–1040. <https://link.springer.com/article/10.1007/s10479-023-05626-6>.
- Zhao, M., Liang, W., Xu, G., Li, Z., et al., 2019. Urban networks of leisure activities: using douban event to measure interaction in the megacity region of the Pearl River delta. In: Ye, X., Liu, X. (Eds.), *Cities as Spatial and Social Networks*. Springer International Publishing, Cham, pp. 165–189. https://doi.org/10.1007/978-3-319-95351-9_10
- Zhao, P., Hu, H., Zeng, L., Chen, J., Ye, X., et al., 2023. Revisiting the gravity laws of inter-city mobility in megacity regions. *Sci. China Earth Sci.* 66 (2), 271–281. <https://doi.org/10.1007/s11430-022-1022-9>
- Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E.P., Zhang, H., Gonzalez, J.E., Stoica, I., 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. *arXiv:2306.05685v4*.
- Zhou, D., 2024. Tourism destination marketing strategy driven by big data. *J. Comput. Methods Sci. Eng.* 24 (4-5), 2593–2609. <https://journals.sagepub.com/doi/abs/10.3233/JCM-247486>.
- Ziakas, V., 2020. Leveraging sport events for tourism development: the event portfolio perspective. *J. Global Sport Manage.* 8 (1), 43–72. <https://doi.org/10.1080/24704067.2020.1731700>
- Ziakas, V., Getz, D., 2021. Event portfolio management: an emerging transdisciplinary field of theory and praxis. *Tourism Manage.* 83, 104233. <https://www.sciencedirect.com/science/article/pii/S026151772030159X>.